Capacity and Limits of Multimodal Remote Sensing: Theoretical Aspects and Automatic Information Theory-Based Image Selection

Saloua Chlaily, *Member, IEEE*, Mauro Dalla Mura, *Senior Member, IEEE*, Jocelyn Chanussot, *Fellow, IEEE*, Christian Jutten, *Fellow, IEEE*, Paolo Gamba, *Fellow, IEEE*, and Andrea Marinoni, *Senior Member, IEEE*

***Abstract*— Although multimodal remote sensing data analysis can strongly improve the characterization of physical phenomena on Earth's surface, nonidealities and estimation imperfections between records and investigation models can limit its actual information extraction ability. In this article, we aim at predicting the maximum information extraction that can be reached when analyzing a given data set. By means of an asymptotic information theory-based approach, we investigate the reliability and accuracy that can be achieved under optimal conditions for multimodal analysis as a function of data statistics and parameters that characterize the multimodal scenario to be addressed. Our approach leads to the definition of two indices that can be easily computed before the actual processing takes place. Moreover, we report in this article how they can be used for operational use in terms of image selection in order to maximize the robustness of the multimodal analysis, as well as to properly design data collection campaigns for understanding and quantifying physical phenomena. Experimental results show the consistency of our approach.**

***Index Terms*— Capacity, data analysis, image selection, multimodal remote sensing, reliability.**

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Saloua Chlaily and Andrea Marinoni are with the Department of Physics and Technology, UiT the Arctic University of Norway, NO-9037 Tromsø, Norway (e-mail: saloua.chlaily@uit.no; andrea.marinoni@uit.no).

Mauro Dalla Mura is with Grenoble Images Parole Signals Automatics Laboratory (GIPSA-lab), Department of Images and Signals, Grenoble Institute of Technology, 38402 Grenoble, France, and also with the Tokyo Tech World Research Hub Initiative (WRHI), School of Computing, Tokyo Institute of Technology, Tokyo 152-8550, Japan (e-mail: mauro.dalla-mura@gipsa-lab.grenoble-inp.fr).

Jocelyn Chanussot is with CNRS, Grenoble INP, GIPSA-lab, University of Grenoble Alpes, 38000 Grenoble, France (e-mail: jocelyn.chanussot@gipsa-lab.grenoble-inp.fr).

Christian Jutten is with the GIPSA-lab, University of Grenoble Alpes, 38000 Grenoble, France, and also with the Institut Universitaire de France, 75005 Paris, France (e-mail: christian.jutten@gipsa-lab.grenoble-inp.fr).

Paolo Gamba is with the Telecommunications and Remote Sensing Laboratory, Department of Electrical, Computer, and Biomedical Engineering, The University of Pavia, 27100 Pavia, Italy (e-mail: paolo.gamba@unipv.it).

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I. Introduction

RECENTLY, multimodal remote sensing has attracted the interest of several scientists, mainly because of the huge potential encompassed by the diversification of the data [1]–[9]. It is expected that multimodal remote sensing enhances the understanding of Earth's surface physical phenomena. Nonetheless, it has been proved, either in remote sensing [4], [5] or in other data science research fields [3], [10], that processing more data and records does not always result in detailed information extraction because of nonidealities, mismatches, and estimation imperfections [5], [11], [12].

This effect is even more evident in a multimodal analysis setup, as the differences in temporal, spatial, spectral, and radiometric resolutions might affect the correct alignment, labeling, and reference of the records to be processed [1]–[3], [5], [8], [9]. For instance, the outcome of the 2009–2010 annual contest on remote sensing data fusion showed that the set of two optical images alone could ensure better change detection in a flooded area than when combined with two SAR images [5]. This can be considered as an example pointing out that increasing the size and diversity of a data set does not always imply an improvement in the accuracy and robustness of the analysis.

Thus, investigations have been conducted to extract accurate and reliable information from a set of heterogeneous records [3], [4]. These studies are instrumental to understand the limits of multimodal remote sensing analysis, as well as to quantify the actual benefits that such an analysis can provide to the characterization of phenomena occurring on Earth's surface. In fact, the effectiveness of multimodal remote sensing information extraction varies depending on the assumptions on feature statistics and the knowledge of the sample distributions. In order to obtain the best information extraction performance from multiple images, it is expected that the multimodal estimation procedure would ideally weight the

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inputs coming from any considered data set, so as to make the best use of the relevant information collected by the different sensors. This would result in an adaptive scheme to achieve optimality in information extraction but might be too complex to implement without a complete knowledge of feature distributions and pixel noise statistics [13].

Understanding the maximum grade of information extraction (i.e., capacity) that can be achieved starting from a data set with specific statistical properties plays a fundamental role in properly assessing the maximum performance of a multimodal analysis scheme. Furthermore, quantifying the reliability of the outcomes that a multimodal investigation architecture can produce is a key factor in evaluating the value of the outputs [6], [11]–[14].

These quantities can have operational use in several aspects of environmental science. In fact, they could enhance the selection of data subsets to achieve maximum information extraction. In other terms, it would be possible to understand which images, features, or classifications are providing information on the considered scene and retrieve the best subset from the remote sensing data pool. At the same time, this result can lead to an enhancement of the system efficiency since only the relevant features and attributes from the different images will be retrieved.

Furthermore, quantifying the capacity and reliability of multimodal data would help in planning campaigns for pointwise relevant field investigation. Indeed, estimating *a priori* the reliability of our analysis on a given set of remotely sensed data over a specific area can help policymakers and end-users in verifying whether further missions (e.g., acquisition of more images or man-driven exploration in a specific region) would be required to characterize the phenomena of interest over the given region. Therefore, a strong improvement in the efficiency of data acquisition strategies (both in terms of financial costs and environmental impact) can be achieved by estimating the maximum information extraction that can be obtained from a new set of data. This last aspect is particularly important when extreme regions (e.g., polar areas, oceans, and tropical forests) are analyzed since they typically require a strong effort in terms of manpower for data acquisition campaigns.

In this article, we define the capacity, i.e., the maximum accuracy, that the data analysis can achieve in a framework for multimodal remote sensing, as a function only of the images' statistics, and independently of the sample distributions. Our approach consists in considering the optimal adaptive weighting of the inputs able to achieve the maximum information extraction. This optimal setup is obtained by an asymptotic investigation based on information theory. The asymptotic analysis enables the derivation of approximate expressions that can be measured before the multimodal processing takes place and provides useful insight into the reliability of any outcomes. Thus, the main contributions of this article are as follows:

1) the introduction of an information theory-based approach to characterize the maximum information extraction performance as a function of data statistics and parameters that characterize the multimodal scenario to be addressed;
2) the definition of two indices to quantify maximum accuracy and reliability of the investigation strategy that can take place over the considered multimodal records;
3) the use of an asymptotic investigation to approximate the expressions of the aforementioned indices so that these quantities can be estimated during preprocessing, while the use of difficult numerical integration and root finding techniques can be avoided.

It is worth emphasizing that, to the best of our knowledge, this work represents the first effort devoted to the understanding of the information extraction process from multimodal remote sensing data, as this subject has not been addressed in these terms by any article in technical literature. As such, in this article, we provide a thorough theoretical walk-through into the system setup, the information theory-based definitions that we have used, and the approximations that we have considered. Moreover, in this article, we provide the evidence of the link between the capacity that we can compute in preprocessing and the actual overall accuracy (OA) that can be obtained by completing the multimodal remote sensing data processing chain.

In the second part of this article, we aim at showing the actual potential of the aforesaid investigation and the proposed two indices for reliability and capacity of multimodal remote sensing data analysis. Especially, we report the results achieved when setting two architectures (based on genetic algorithm and branch-and-bound (BB) scheme) aiming at the capacity maximization to automatically select at the data level the relevant information in the considered multimodal data sets. The achievements obtained when introducing the proposed image selection methods provide a solid overview of the actual impact that our information theory-based approach can have in unveiling the details of capacity and limits of information extraction in multimodal remote sensing applications, as well as in boosting the effective interpretation of the significant characteristics of the scenes.

The remainder of this article is organized as follows. Section II reports the derivation of the capacity and reliability metrics. Section II-A introduces the proposed model for the multimodal data system and its motivation, and the details of the proposed information theory-based analysis are delivered in Sections II-B–II-D with computation details in the Appendix. In Section III, we introduce the methods for automatic image selection based on the asymptotic information indices that we have proposed. Then, Section IV delivers experimental performance results to show the consistency of our approach. Finally, Section V delivers our final remarks and some ideas on future research steps.

For notational convenience, random scalars are denoted by lower case letters, e.g., z. Underlined lower case letters designate random vectors, e.g., $\underline{z}$. The hat is used to denote estimates, e.g., $\hat{\underline{z}}$ refers to the estimate of $\underline{z}$. Double underlined upper case letters refer to matrices, e.g., $\underline{\underline{A}}$, $\mathbb{E}[\cdot]$ is the expectation operator, $\underline{z}^T$ identifies the transpose of $\underline{z}$, and $\underline{\underline{I}}_a$ is the $a \times a$ identity matrix. Finally, $|\underline{\underline{A}}|$ is the determinant of matrix $\underline{\underline{A}}$.

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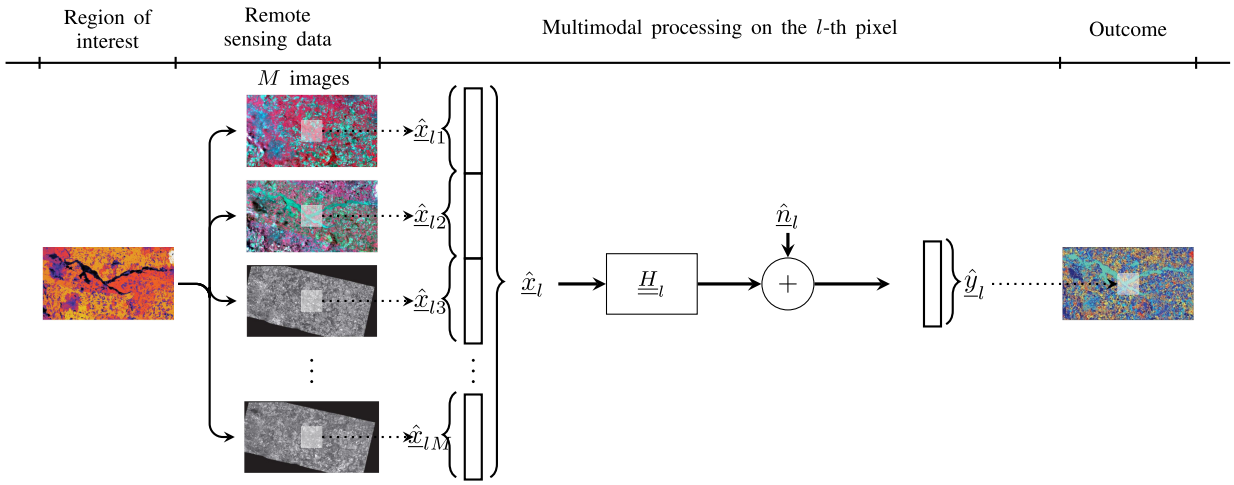


Fig. 1. General multimodal remote sensing analysis scheme that has been considered in this work. The multimodal processing is shown for the lth pixel represented by gray squares in each remotely sensed image and also in the outcome of the analysis. Graphics are taken from the multimodal data set delivered in the IEEE GRSS Data Fusion Contest 2009–2010 [5].

II. Capacity and Reliability of Multimodal Remote Sensing

A. System Model

For the sake of clarity, we consider multimodal data fusion at the decision level. In decision-level fusion, the estimates are drawn separately from each remotely sensed image or image set [4]. Then, the multimodal analysis combines these estimates to obtain more accurate and robust results. The results of this article can be easily extended to other levels of fusion, i.e., feature level and data level [3], [4].

Let M be the number of the remotely sensed images to be processed and L the numbers of pixels in the reference image. We denote by $\hat{\underline{x}}_{lm} = [\hat{x}_{lmr}]_{r=1,\ldots,R}$ the set of decision outcomes of the lth pixel drawn from the mth image used to feed the multimodal analysis. The value and meaning of R and $\hat{\underline{x}}_{lm}$ would vary according to the final purpose of the considered analysis [1], [2], [6], [15]. For instance, in an unmixing framework, R would be the number of end-members in the region, and $\hat{x}_{lmr}$ would be the abundance of the rth end-member within the lth pixel. In a supervised (unsupervised) classification scheme, R would be the number of classes (thematic clusters), and $\hat{x}_{lmr}$ would be the probability to be assigned to the rth class of the lth pixel in the region (thematic cluster). Finally, in target detection, R would be set to 1, and $\hat{x}_{lm}$ would then identify the likelihood of occurrence of the given target in the lth pixel.

We write the estimates drawn on the lth pixel of the region of interest out of the mth image as $\hat{\underline{x}}_{lm} = f_{lm}(\underline{\underline{Y}})$, where $\underline{\underline{Y}} = [\underline{y}_l]_{l=1,\ldots,L}$, with $\underline{y}_l = [y_{lr}]_{r=1,\ldots,R}$, is the reference image. By the reference image, we mean the information that we aim to recover on the observed scene. We can assume, without losing generality, that y_{lr} might live in a Q-level quantized $[0, 1]$ interval, i.e., $y_{lr} \in \{i/(Q-1)\}_{i=0,1,\ldots,Q-1}$. The f_{lm} functions identify different manipulations of the features acquired in the mth image for properly addressing the mapping of different resolutions in terms of ground cover, geometry, radiometry, and spectral characterization (e.g., by filtering, convolving, integrating, or inferring).

Multimodal remote sensing combines the estimates $\hat{\underline{x}}_l$ in order to obtain a stable assessment of the actual $\underline{y}_l$ distribution for each pixel in the ground-truth image. Especially, $\hat{\underline{y}}_l \in \mathbb{R}^R$ is the outcome of the multimodal analysis, obtained by combining the estimates $\hat{\underline{x}}_l = [\hat{\underline{x}}_{lm}]_{m=1,\ldots,M} \in \mathbb{R}^{RM}$

$$\hat{\underline{y}}_l = \underline{\underline{H}}_l \hat{\underline{x}}_l + \underline{n}_l \tag{1}$$

where $\underline{n}_l \in \mathbb{R}^R$ and $\underline{\underline{H}}_l \in \mathbb{R}^{R \times RM}$ model the imperfections and undesired mismatches generated through the estimation process. Although it may be more appropriate to model the effects of mismatches on multimodal remote sensing as nonlinear, a linear model enables the derivation of simple analytic expressions that are more convenient to implement, analyze, and interpret. Finally, we can assume that each element of $\hat{\underline{x}}_l$ lives in the Q-level quantized $[0, 1]$ interval as the aforesaid elements of $\underline{\underline{Y}}$. Fig. 1 reports a schematic workflow of the aforesaid model.

We consider every element $\underline{n}_l$ to be zero-mean Gaussian distributed. Moreover, to simplify the calculations, we state that $\mathbb{E}[\underline{n}_l^T \underline{n}_l] = N_0 \underline{\underline{I}}_R \ \forall l$, where $N_0 \in \mathbb{R}_+$ is the noise variance. It is noteworthy that the model is still valid even if $\mathbb{E}[\underline{n}_l^T \underline{n}_l] \neq N_0 \underline{\underline{I}}_R$ since it suffices to multiply $\hat{\underline{y}}_l$ in (1) by the inverse of the noise covariance matrix yielding to the channel matrix $(\mathbb{E}[\underline{n}_l^T \underline{n}_l])^{-1} \underline{\underline{H}}_l$. We will also assume $\underline{\underline{H}}_l$ to be stationary and ergodic as well as to be composed of independent and identically distributed (i.i.d.) Gaussian random variables with zero mean. Moreover, we can assume $\underline{\underline{H}}_l$ to be slowly varying along the pixels of the reference map.

It is worth noting that these assumptions on the statistical properties of the terms in (1) are widely employed in the technical literature. Especially, the assumptions of stationarity and ergodicity of the system either in terms of data statistics (e.g., [1], [16], [17], and [18] ch. 3) and processing approach (e.g., [2], [18]–[21] ch. 16) are implied and/or drawn in order to maximize the tradeoff between precision and efficiency of the remote sensing data analysis frameworks. It is also true that in some specific application scenarios, these assumptions (especially, ergodicity) might not stand, particularly when

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specific applications with *a priori* known statistical properties in dynamic environments are targeted (see [22]–[24]). Nonetheless, when no reliable *a priori* statistical knowledge can be specifically addressed and/or when facing the ambition to derive a general description for a multipurpose data analysis scheme (as in the case of this work), the assumptions in (1) are able to provide a solid bedrock to achieve a good generalization of the system.

Furthermore, the statistical properties of $\underline{\underline{H}}_l$ and $\underline{n}_l$ might indeed follow several non-Gaussian distributions (e.g., Poisson-like and exponential), as nonidealities can show up along the multimodal remote sensing framework in several ways (e.g., detectors affected mainly by photon noise or images affected by striping effect). This article aims at focusing on the definition of the optimal information extraction performance of a multimodal remote sensing data analysis framework in the case of most deteriorating imperfections. As such, it has been proven in [25] that the Gaussian distributed imperfections would lead to a strong degradation of the information retrieval results compared with other imperfections.

It is possible to reshape the model in (1) through a singular value decomposition of $\underline{\underline{H}}_l$ [26] so that describing its information extraction properties can be facilitated. Let us define $\underline{\underline{W}}_l = \underline{\underline{H}}_l \underline{\underline{H}}_l^T \in \mathbb{R}^{R\times R}$. $\underline{\underline{W}}_l$ can have at most R nonzero eigenvalues as it is a Wishart matrix by definition [27]. Exploiting this property, we can decompose $\underline{\underline{H}}_l$ by means of a singular value decomposition as $\underline{\underline{H}}_l = \underline{\underline{U}}_l \underline{\underline{\Lambda}}_l \underline{\underline{V}}_l^T$, where $\underline{\underline{U}}_l \in \mathbb{R}^{R\times R}$ and $\underline{\underline{V}}_l \in \mathbb{R}^{RM\times RM}$ are unitary matrices, i.e., $\underline{\underline{U}}_l \underline{\underline{U}}_l^T = \underline{\underline{I}}_R$ and $\underline{\underline{V}}_l \underline{\underline{V}}_l^T = \underline{\underline{I}}_{RM}$. Moreover, $\underline{\underline{\Lambda}}_l \in \mathbb{R}^{R\times RM}$ has nonzero elements only on its main diagonal, i.e., $\underline{\underline{\Lambda}}_{l_{rr}} = \sqrt{\lambda_{lr}}$ ($r \in \{1,\ldots,R\}$) and $\underline{\underline{\Lambda}}_{l_{rk}} = 0$ for $k \neq r$ ($k \in \{1,\ldots,MR\}$). $\sqrt{\lambda_{lr}}$ denotes the nonnegative square roots of the eigenvalues of $\underline{\underline{W}}_l$. Hence, by applying the aforesaid expression of $\underline{\underline{H}}_l$, we obtain the following alternative expression of (1):

$$\underline{\tilde{y}}_l = \underline{\underline{\Lambda}}_l \underline{\tilde{x}}_l + \underline{\tilde{n}}_l \tag{2}$$

where $\underline{\tilde{y}}_l = \underline{\underline{U}}_l^T \underline{\hat{y}}$, $\underline{\tilde{x}}_l = \underline{\underline{V}}_l^T \underline{\hat{x}}_l$, and $\underline{\tilde{n}}_l = \underline{\underline{U}}_l^T \underline{n}_l$. It is important to emphasize that since $\underline{\underline{U}}_l$ and $\underline{\underline{V}}_l$ are unitary, the statistical properties of $\underline{\hat{x}}_l$ and $\underline{n}_l$ are maintained by $\underline{\tilde{x}}_l$ and $\underline{\tilde{n}}_l$, respectively. Namely, $\underline{n}_l$ and $\underline{\tilde{n}}_l$ have the same distribution, and $\mathbb{E}[\underline{\hat{x}}_l^T \underline{\hat{x}}_l] = \mathbb{E}[\underline{\tilde{x}}_l^T \underline{\tilde{x}}_l]$.

Before introducing the method used to derive the capacity and reliability metrics, let us define and summarize in Table I some additional quantities that are crucial for our analysis.

B. Information Theory-Based Analysis

The final purpose of an estimation scheme is to achieve the minimization of the variance between the target parameters and the produced estimates. This can be expressed, according to the Cramer–Rao bound, in terms of the maximization of the mutual information between those two variables [11], [13], [28]. In estimation theory, this translates into quantifying how much of the target parameters can be reliably described by considering the set of observed variables. Hence, given θ the variable to be estimated and $\hat{\theta}$ its estimate, the mutual information can be expressed as $I(\theta,\hat{\theta}) = \sum_{\theta\in\Theta}\sum_{\hat{\theta}\in\hat{\Theta}} p(\theta,\hat{\theta}) \log((p(\theta,\hat{\theta}))/(p(\theta)p(\hat{\theta})))$, where $p(\theta,\hat{\theta})$ identifies the joint probability of θ and $\hat{\theta}$, while $p(z)$ reports the marginal probability of z. Furthermore, the upper bound of $I(\theta,\hat{\theta})$ is called capacity, i.e., $\mathcal{C} = \sup_{p(\theta)} I(\theta,\hat{\theta})$. Therefore, the capacity determines the maximum value of information on θ that can be reliably extracted by the considered estimation system that provides $\hat{\theta}$ [13].

TABLE I
TABLE OF SYMBOLS

$\hat{P}_{lr} = \frac{1}{M}\sum_{m=1}^{M} \hat{x}_{lmr}^2$	Average power of r-th component within the l-th pixel
$\hat{P}_l = \frac{1}{R}\sum_{r=1}^{R} \hat{P}_{lr}$	Average power of l-th pixel
$\hat{P} = \frac{1}{L}\sum_{l=1}^{L} \hat{P}_l$	Average power of all pixels
$\bar{\zeta} = \frac{\hat{P}}{RLN_0}$	Average SNR on the M images of all pixels
$\zeta_{lr} = \lambda_{lr}\frac{\hat{P}_{lr}}{N_0}$	SNR of the r-th component within the l-th pixel

In this article, we are interested in investigating the capacity of a multimodal remote sensing system. Thus, we must compute the upper bound of the mutual information between the estimate of the reference image $\underline{\underline{\hat{Y}}}$ based on the fusion of local estimates for all the M images and the set of estimates for each pixel $\underline{\underline{\hat{X}}} = [\underline{\hat{x}}_l]_{l=1,\ldots,L}$. Given the expressions in (1) and (2), it should not be surprising that the capacity for a multimodal remote sensing system depends on the images' statistical characteristics as well as on the estimator performance, modeled by $\underline{n}$ and $\underline{\underline{H}}$. Nonetheless, when investigating the optimal maximum information extraction behavior, the assumptions that we have considered in the previous section are useful to find an analytical expression for the capacity of the multimodal system that is independent of the statistical distributions of the images.

Using the Gaussian assumptions (further details are given in Appendix A), we define the capacity of the system in (1) as follows:

$$\mathcal{C} = \max_{\hat{P}_l} \sum_{l=1}^{L} \mathbb{E}_{\underline{\underline{H}}_l}\left[\log_Q\left|\underline{\underline{I}}_R + \frac{\underline{\underline{H}}_l \mathbb{E}[\underline{\hat{x}}_l \underline{\hat{x}}_l^T] \underline{\underline{H}}_l^T}{N_0}\right|\right] \tag{3}$$

where the maximum is over the average power of the lth pixel, $\hat{P}_l = \mathbb{E}[\underline{\hat{x}}_l^T \underline{\hat{x}}_l]$, $\log_Q x = (\ln x/\ln Q)$, and Q refers to the quantization levels of $\underline{\hat{x}}_l$. The expression in (2) allows to simplify this equation to (see Appendix A)

$$\mathcal{C} = \max_{\hat{P}_{lr}} \sum_{l=1}^{L}\sum_{r=1}^{R} \mathbb{E}_{\zeta_{lr}}\left[\log_Q\left(1 + \frac{\zeta_{lr}\hat{P}_{lr}}{\hat{P}}\right)\right] \quad \text{s.t.} \ \frac{1}{LR}\sum_{l,r}\hat{P}_{lr} = \hat{P} \tag{4}$$

where the maximum is taken over $\hat{P}_{lr}$ the average power (on the M images) of the rth component within the lth pixel, and ζ_{lr} is the SNR of the rth component within the lth pixel (see Table I).

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The maximum information extraction can be achieved by an optimal power allocation that consists in attributing high power to more reliable estimates and, conversely, low, if not zero, power to unreliable estimates while guaranteeing that the power constraint (i.e., $(1/LR)\sum_{l=1}^{L}\sum_{r=1}^{R}\hat{P}_{lr}=\hat{P}$) is still satisfied [13]. The following sections provide more details on the terms of this maximization.

C. Reliability Metric

The maximization in (4) is obtained by setting $\hat{P}_{lr}$ using a Lagrange multiplier as follows [29]:

$$\frac{\hat{P}_{lr}}{\hat{P}}=\begin{cases}Z_0^{-1}-\zeta_{lr}^{-1}, & \text{if } \zeta_{lr}\geq Z_0\\ 0, & \text{if } \zeta_{lr}<Z_0\end{cases} \tag{5}$$

where $r\in\{1,\ldots,R\}$, and Z_0 is the Lagrange multiplier.

According to the above equation, Z_0 can be considered as a cutoff value above which the retrieved estimates can be taken into account. In fact, if the estimates in $\underline{\hat{x}}_l$ show a signal-to-noise ratio (SNR) (defined according to the meaning of ζ_{lr}) smaller than Z_0, they should be discarded to ensure the maximum information extraction. Thus, Z_0 works as a metric that assesses the reliability of the estimates to provide the best understanding of the considered region. We can, therefore, ultimately use Z_0 to quantify the maximum confidence that we can achieve from the given multimodal framework [13].

Since the value of $\hat{P}_{lr}$ varies with respect to ζ_{lr}, we write the discrete power constraint, i.e., $((\sum_l\sum_r\hat{P}_{lr})/LR)=\hat{P}$, as a continuous power constraint $\int\hat{P}_{lr}\,p_{\zeta_{lr}}(\zeta_{lr})\mathrm{d}\zeta_{lr}=\hat{P}$. In order to derive the expression of Z_0, we substitute $\hat{P}_{lr}$ in the continuous power constraint by its definition in (5). Hence, Z_0 fulfills

$$\int_{Z_0}^{\infty}\left(Z_0^{-1}-\zeta_{lr}^{-1}\right)p_{\zeta_{lr}}(\zeta_{lr})\mathrm{d}\zeta_{lr}=1 \tag{6}$$

where $p_{\zeta_{lr}}(\zeta_{lr})$ is the probability distribution function of ζ_{lr}. Let us now consider the general distribution of the ζ_{lr} values over the whole data set, i.e., $p_{\zeta}(\zeta)$. Thus, following the results in [14], (6) turns into

$$\int_{Z_0}^{\infty}\left(\frac{1}{Z_0}-\frac{1}{\zeta}\right)p_{\zeta}(\zeta)\mathrm{d}\zeta=1. \tag{7}$$

Moreover, it is possible to define $p_{\zeta}(\zeta)$ by means of the properties of $\underline{\underline{W}}_l$. Indeed, recalling that $\underline{\underline{W}}_l$ is a Wishart distributed matrix [30], $p_{\zeta}(\zeta)$ can be written as using that $\lambda=\zeta/\bar{\zeta}$

$$p_{\zeta}(\zeta)=\bar{\zeta}^{-1}p_{\lambda}(\zeta/\bar{\zeta}) \tag{8}$$

where $\bar{\zeta}=(\hat{P}/RLN_0)$ is the average SNR, on all images, of all pixels (see Table I). $p_{\lambda}(\zeta/\bar{\zeta})$ represents the probability distribution function of an unordered eigenvalue of a Wishart distributed matrix [13], [14], [26], [31], that is,

$$p_{\lambda}(u)=\frac{e^{-u}u^{R(M-1)}}{R}\sum_{r=1}^{R}\kappa_r\left[\mathcal{L}_{r-1}^{R(M-1)}(u)\right]^2 \tag{9}$$

being $\mathcal{L}_s^t(u)=(1/s!)e^u u^{-t}(\mathrm{d}^s/\mathrm{d}u^s)(e^{-u}u^{s+t})$ the sth order Laguerre polynomial and $\kappa_r=((r-1)!/(RM-R+r-1)!)$.

It is possible to prove that Z_0 is living in $[0,1]$, $\lim_{\bar{\zeta}\to 0}Z_0=0$, and $\lim_{\bar{\zeta}\to\infty}Z_0=1$ [14]. Moreover, (7) [and thus (4)] has a unique solution in Z_0. Hence, there is only one value of Z_0 able to provide the maximization of the information extraction in the multimodal system in (1). In Appendix B, we detail the proof of the uniqueness of Z_0.

The expression of Z_0 can be obtained by working on the algebraic properties of the terms involved in (7), as provided in Appendix C. In fact, (7) can be written, with $\nu=Z_0/\bar{\zeta}$, as

$$\sum_{r=1}^{R}\kappa_r\sum_{j_1=0}^{r-1}\sum_{j_2=0}^{r-1}\frac{(-1)^{j_1+j_2}}{j_1!j_2!}J_1J_2\frac{\Gamma(\beta_1,\nu)-\nu\Gamma(\beta_2,\nu)}{\nu}=R\bar{\zeta} \tag{10}$$

where $J_1=\binom{R(M-1)+r-1}{r-1-j_1}$ and $J_2=\binom{R(M-1)+r-1}{r-1-j_2}$, $\beta_2=R(M-1)+j_1+j_2$, $\beta_1=\beta_2+1$, and $\Gamma(s,t)$ is the complementary incomplete Gamma function, i.e., $\Gamma(s,t)=\int_t^{\infty}u^{s-1}e^{-u}\mathrm{d}u$. The above equation can be solved for a unique value of the cutoff Z_0 by means of iterative numerical search techniques. Nevertheless, it is possible to obtain an approximate estimate of Z_0 by means of a small perturbation procedure applied to ν in (10). Let us consider $\alpha(t)=\sum_{r=1}^{R}\kappa_r\sum_{j_1=0}^{r-1}\sum_{j_2=0}^{r-1}(((-1)^{j_1+j_2})/(j_1!j_2!))J_1\ J_2\ t!$. Hence, it is possible to write the approximate of Z_0 as

$$Z_0\approx\widehat{Z_0}=\frac{\bar{\zeta}\alpha(\beta_2)}{R\bar{\zeta}+\alpha(\beta_2-1)}. \tag{11}$$

It is worth noting that as the first result of the aforesaid definition, (11) becomes an algebraic equation for $\bar{\zeta}\gg 1$, i.e., $\lim_{\bar{\zeta}\to\infty}\widehat{Z_0}=(1/R)\alpha(\beta_2)=1$.

Now, to understand how the reliability metric reacts to various setups of multimodal remote sensing data analysis framework, using (11), we report, in Fig. 2, the variation of $\widehat{Z_0}$ as a function of $\bar{\zeta}$ and for different values of M and R. We remark that $\widehat{Z_0}$ trends change significantly for different setups. For a given M and R, $\widehat{Z_0}$ increases with the average SNR, which is expected since the extracted information from the images gets more precise. Moreover, while comparing the subfigures, we notice that the curves are shifted to the right when the number of classes/thematic clusters R increases, showing that the reliability decreases when R increases. For example, the optimal information extraction is reached with maximum reliability (when $M=2$) for $\bar{\zeta}$ equal to 20, 27, 35, and more than 40 dB for $R=1,\ldots,4$, respectively. In fact, by increasing the number of classes/thematic clusters, the complexity of the analysis increases. In this case, more data and better SNR are required to extract further details about the observed region. Finally, for a given $\bar{\zeta}$ and R, the reliability increases with the number of images since additional information can be extracted. However, increasing the number of images is not always relevant. For instance, the curves for $M>35$ collapse when R equals 3 and 4. In these cases, the optimal information extraction is achieved with fewer images, and additional data do not provide any novelty.

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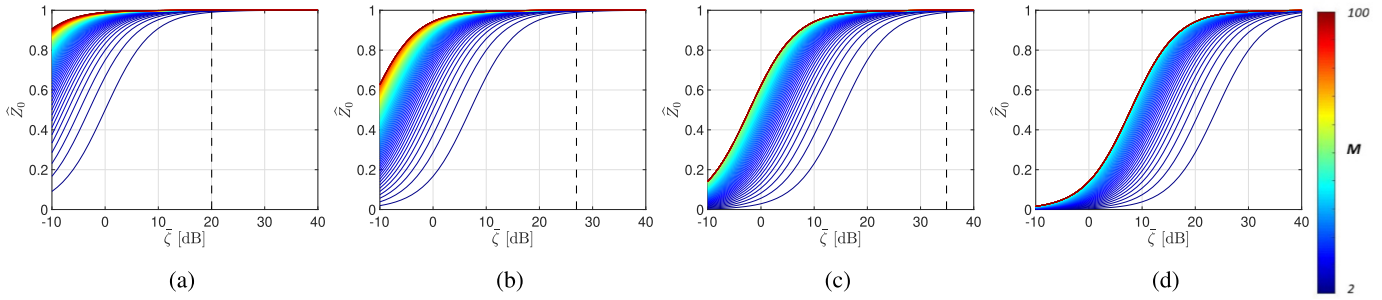


Fig. 2. Reliability estimates obtained as in (11) for different settings of a multimodal remote sensing analysis framework. The color of the curves identifies the number of images M that are considered, according to the color map on the right-hand side of the figure. The number of classes/thematic clusters R varies from 1 to 4 in (a)–(d), respectively. The trends of $\widehat{Z}_0$ are reported as a function of the SNR $\bar{\zeta}$. The dashed lines identify the minimum value of $\bar{\zeta}$ for which the maximum reliability is achieved by processing two images for different number of classes/thematic clusters R. (a) $R = 1$. (b) $R = 2$. (c) $R = 3$. (d) $R = 4$.

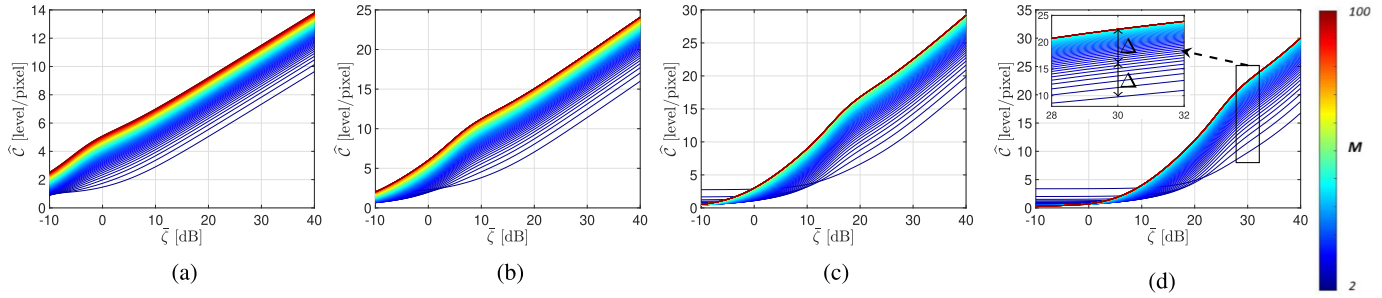


Fig. 3. Capacity estimates according to (13) for different settings of a multimodal remote sensing analysis framework. The same legend as in Fig. 2 applies here. For better visibility, the scale of the vertical axis is different for each subfigure. (a) $R = 1$. (b) $R = 2$. (c) $R = 3$. (d) $R = 4$.

D. Capacity Metric

Let us focus now on the maximization of (4). Given the optimization settings in (5) and the aforementioned results on the cutoff value Z_0, the capacity of a multimodal system as in (1) can be written as follows:

$$\mathcal{C} = LR \int_{Z_0}^{\infty} \log_Q \left(\frac{\zeta}{Z_0} \right) p_\zeta(\zeta) \mathrm{d}\zeta \tag{12}$$

where $p_\zeta(\zeta)$ is the probability distribution function of any eigenvalue in the system as in (8). It is possible to write an approximation of (12) in a consistent way with respect to (10), following [14], [32]:

$$\mathcal{C} \approx \widehat{\mathcal{C}} = \sum_{r=1}^{R} \kappa_r \sum_{j_1=0}^{r-1} \sum_{j_2=0}^{r-1} \frac{(-1)^{j_1+j_2}}{j_1! j_2!} J_1 J_2 \mathcal{B}_{\beta_1}(\nu). \tag{13}$$

where

$$\mathcal{B}_t(\nu) = (t-1)! \left[E_1(\nu) + \sum_{k=1}^{t-1} \mathcal{P}_k(\nu)/k \right] \tag{14}$$

where $E_1(\nu) = \int_\nu^\infty e^{-u} u^{-1} \mathrm{d}u$ is the exponential integral function, whereas $\mathcal{P}_k(\nu) = e^{-\nu} \sum_{m=0}^{k-1} \nu^m / m!$ is the kth order Poisson sum. Moreover, let us recall that $\beta_1 = R(M-1) + j_1 + j_2 + 1$.

$\mathcal{C}$ reports the number of quantization levels that can be reliably analyzed and processed over the given set of multimodal images per areal unit. Hence, it provides a quantification of the precision that can be reached when analyzing the considered region, i.e., the degree of understanding that can be extracted on the area of interest. This approximation $\widehat{\mathcal{C}}$ in (13) describes the level of detail reachable if the given multimodal analysis framework works optimally. $\widehat{\mathcal{C}}$ is a function of the local SNR computed over the images and estimates. Hence, by computing the SNR of the given remote sensing data pool, the information extraction limits for any multimodal data set can be calculated.

In Fig. 3, we show capacity curves as function of the average SNR $\bar{\zeta}$ according to (13). The different subfigures correspond to a different number of classes/thematic clusters R, and the curves' colors refer to the number of used images M. Note that the scale of the vertical axis is different for each subfigure. It is not surprising that the precision of the optimal information extraction[1] increases with the number of classes/thematic clusters R, with the average SNR and with the number of images M. In fact, in these cases, further details about the region of interest can be obtained. However, the gain in capacity decreases by increasing the number of images until it is saturated, and no improvement can be obtained, as can be seen in the zoomed-in part of Fig. 3(d). In fact, compared with two images, 12 images provide a capacity gain of Δ, which represents half of the gain obtained by adding 98 images. Moreover, we notice that curves with low M deliver a higher value of capacity, especially for higher R and lower $\bar{\zeta}$. Low values of M correspond to the case where a small amount of images is considered. This result might seem counterintuitive since it is commonly presumed that in the case of a noisy data set, more data ensure better results. However, for the data fusion to be fruitful, the multimodal analysis should be

[1]It is measured in the number of levels that can be reliably solved for an areal unit.

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able to capture the underlying structure of the images. This task becomes difficult when increasing the size of the data set and even more complicated when the considered data set is noisy. The capacity metric is able to determine the threshold on the average SNR over which more data actually imply better results. Note that the value of such a threshold increases with the complexity of the analysis (i.e., with R).

III. Automatic Image Selection

As mentioned earlier, the metrics derived in Section II can be used to perform a selection of the relevant contributions within the considered data sets at data, feature, and decision levels. In this article, we focus on building a framework for automatic image selection, i.e., on designing an architecture that selects the most informative scenes in the considered data set in order to perform the best information extraction. It is worth emphasizing that this can be considered as a straightforward application of the notions that we addressed in Section II. Indeed, other applications could be also considered. Especially, future works will then be dedicated to the use of $\widehat{Z}$ and $\widehat{\mathcal{C}}$ for enhanced feature selection (FS) and ensemble learning when combining the decisions drawn from a set of different classifiers on the same set of records.

Selecting the most informative subset of images identifies a complicated optimization problem, which can be generally described as follows:

$$\arg\max_{x\in\mathcal{D}} \ \widehat{\mathcal{C}}(x) \tag{15}$$

where $\mathcal{D}$ identifies the set of all possible combinations of the images in the data set and $\widehat{\mathcal{C}}(x)$ is the value obtained when all the quantities in (13) are computed over the subset x.

Indeed, the goal that we would like to achieve directly translates into optimizing the number of images to be considered, as well as the actual composition of the selected subset. As such, it is possible to expect that the optimization problem that we must face is nonlinear and eventually nonconvex [30], [33]. Therefore, in general, this problem might have multiple local solutions. The number of such local solutions is not known *a priori*, and the quality of global and local solutions may differ substantially [33].

Thus, it is not possible to directly use standard optimization strategies to solve this problem, and decision models can be very difficult to apply [30], [33]. In fact, classic local search methods might be very sensitive to the initial state of the optimization procedure, which can strongly affect the quality of the search process. Hence, a global search method must be considered. Several global search methods have been introduced in data analysis and remote sensing processing (see, for instance, [6], [12], [30], [33]–[36]). In particular, global search methods can be categorized as belonging to two main classes: exact methods and heuristic methods.

Exact methods aim at directly solving the problem by exhaustively enumerating all possible solutions [37], [38]; visiting all stationary points of the objective function leading to the list of all optima [39]; replacing the problem through a successive refinement process, by a sequence of relaxed subproblems that are easier to solve [38]; allowing a stochastic description of the modeled function/class so that the characteristics of problem/instance are adaptively estimated and updated during optimization [40]; random sampling within the feasible sets to perform adaptive stochastic search [37]; and approximating the level sets of the objective function to determine its supremum [41].

A different approach to achieve global optimization is represented by heuristic algorithms. This family of methods tackles the nonlinearities involved in the feasible set by randomly choosing multiple initial states, so as to trigger several search processes and choose the best solution [37]; starting from an initial solution, performing a set of steps that lead to a new solution inside the neighborhood of the current one while forbidding to move to points already visited in the search space (e.g., tabu methods) [42]; investigating the convexity characteristics of the feasible sets and then performing direct sampling to reach global optimization [43]; transforming the objective function into a smoother function with easily calculated local minimizers and then using a local minimization procedure to trace all minimizers back to the original function [44]; and building auxiliary functions to assist the searching procedure (e.g., tunneling, deflation, and filled function approaches) [45].

In order to perform a solid comparison of the benefits and drawbacks of these algorithms when applied to the maximization of the information extraction, we chose to consider one representative for each of these two families of optimization methods. Especially, we implemented image selection using a BB approach and genetic optimization.

BB algorithms can be considered as strategies based upon adaptive partition, sampling, and bounding procedures to solve global optimization. These operations are applied iteratively to the collection of active subsets within the feasible set. BB algorithms are, therefore, exact methods that typically rely on some *a priori* structural knowledge about the problem, such as how rapidly each function can vary (e.g., the knowledge of a suitable overall Lipschitz constant). This general methodology is applicable to a large number of global optimization problems (such as concave minimization or reverse convex programs [37], [41]). It can be considered as a valid candidate for the practical implementation of the automatic image selection that we would like to achieve.

When applying BB to solve the problem in (15), the overlapping version of BB algorithm (OBB) [46] can be properly applied. To this aim, the bounds of the search procedure are computed by means of nonconvex estimators of the $\widehat{\mathcal{C}}(x)$ function for the considered combination of images x. In practice, the feasible set $\mathcal{D}$ of all the image combinations is iteratively covered by focusing and refining balls $\mathcal{B}$ (i.e., subspaces of $\mathcal{D}$ that are covering multidimensional regions equidistant from their centroid) that are partially overlapping in the space induced by $\mathcal{D}$. The radius of $\mathcal{B}$ is adaptively but monotonically decreased with the iteration progress. This step achieves branching, while the bounding phase is performed by computing lower and upper bounds of $\widehat{\mathcal{C}}(x)$ over the subregion $\mathcal{B}$ as $\max\{0, \widehat{\mathcal{C}}(x_{\mathcal{B}}) \mp L_{\widehat{\mathcal{C}}}(\mathcal{B})\delta_{\mathcal{B}}\}$, where $x_{\mathcal{B}}$ is the closest point of $\mathcal{D}$ to the centroid of $\mathcal{B}$, $\delta_{\mathcal{B}}$ is its distance to the optimization point x, and $L_{\widehat{\mathcal{C}}}(\mathcal{B})$ is the Lipschitz constant of $\widehat{\mathcal{C}}(x)$ over $\mathcal{B}$.

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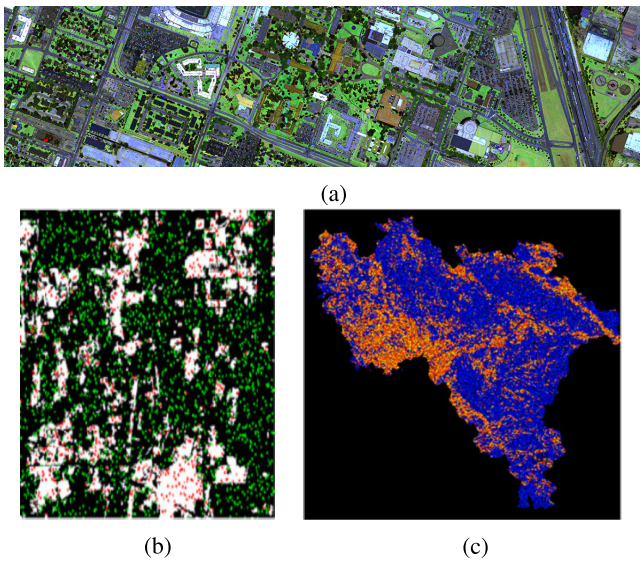


Fig. 4. Few samples of the test data sets. (a) IEEE GRSS Data Fusion Contest 2018 data. (b) Beijing data set. (c) Pavia air quality, glycated hemoglobin, and BMI data set.

On the other hand, genetic algorithms (GAs) heuristically simulate biological evolution [37], [40]. Based on a population of candidate solution points [in (15), $\mathcal{D}$], an adaptive search procedure is applied. The poorer solutions are dropped by a competitive selection, while the remaining pool of candidates with higher values is then recombined with other solutions by swapping components. Based on diverse evolutionary criteria, a variety of deterministic and stochastic algorithms can be constructed. Hence, these architectures can be used to solve the proposed problem by properly defining its structural requirements [40], [42].

It is worth pointing out that a detailed explanation of the theoretical aspects of the optimization schemes employed in this work is beyond the scope of this article. For additional information on the OOB and GA approaches, the reader is encouraged to refer to [46] and [40], respectively.

IV. Experimental Results

The proposed approach has been tested on multiple data sets of remotely sensed records, acquired by optical and synthetic aperture radar (SAR) sensors.

A. Test Data Sets

In this article, we have tested our approach on three different multimodal scenarios, from data sets remotely acquired by multiple selected sensors to multitemporal stacks of remote sensing records and to a collection of data from multiple sources, spaceborne, and *in situ* (see Fig. 4).

First, we considered the data set used for the IEEE GRSS Data Fusion Contest 2018, denoted as DFC2018 in the following [47]. It consists of a collection of remotely sensed data acquired over the University of Houston, TX, USA, on February 16, 2017, covering the University of Houston campus and its surrounding areas. The remote sensing records are a multispectral-LiDAR point cloud data at 1550, 1064, and 532 nm; a hyperspectral data set covering the 380–1050 nm spectral range with 48 bands at a 1-m ground sample distance (GSD); and a very high-resolution RGB imagery at a 5-cm GSD. 20 classes have been identified. Thus, the values of (R, M) according to the notation in the previous section have been set to $(20, 3)$. It is noteworthy that DFC18 used in our analysis consists only of the training data set in [47]. As such, our results cannot be compared with other publications that use the test data set.

Then, we considered a data set of 53 SAR images acquired over Beijing, People's Republic of China, by the ESA Sentinel-1 sensor from February 2015 to March 2016 [48]. Each scene consists of 3265×2448 VV and VH intensity records, where V and H identify the orientation of the polarization (vertical and horizontal, respectively). These records are used to identify regions that could be associated with urban and nonurban classes through time. Therefore, the values of R and M were set to 2 and 53, respectively.

Finally, we took into account a data set that results from gathering together multispectral records collected by the Landsat over the Province of Pavia, Italy, from 2009 to 2014 and yearly and municipality-aggregated glycated hemoglobin and body mass index (BMI) values collected out of a group of 1084 patients affected by diabetes living in the area during the same time interval [15], [49]. Especially, we collected 35 Landsat images, each consisting of 2800×2800 pixels at 30-m spatial resolution. Only the inverse of the thermal infrared band of each Landsat image was used to retrieve the quantity of black particulate concentrated over the area [49]. Especially, five air quality classes (designed according to the limits and specifics provided by the local environmental protection agency to determine the risk of diabetes-related complication onsets, validated by means of black particulate matter sensing ground stations and clinical records [49]) have been considered, i.e., $R = 5$. Moreover, the number of images (M) was set to 35.

B. Validation of $\widehat{Z_0}$ and $\widehat{\mathcal{C}}$

Fig. 5 shows the results in terms of (a) reliability and (b) capacity for these data sets. We computed the trends of $\widehat{Z_0}$ and $\widehat{\mathcal{C}}$ as a function of $\bar{\zeta}$ according to (11) and (13), respectively, and for different values of R and M in the data set, i.e., the number of classes and images considered in each data set. These results are displayed in solid lines. Then, we computed the exact Z_0 by searching for the value that maximizes (7) and the exact value of $\mathcal{C}$ according to (12). These results are shown as stars. Moreover, in order to investigate the actual effectiveness of our approach, we added white Gaussian noise to the records and then computed Z_0 and $\mathcal{C}$ as in (7) and (12) for different values of $\bar{\zeta}$ (dashed lines). These tests would help us to understand and quantify the actual ability of our approach to approximate the maximum values of reliability and capacity that a multimodal system can reach.

For all the data sets, the approximations (11) and (13) are able to top the actual values of Z_0 and $\mathcal{C}$ that have been computed. This result is compliant with the assumptions

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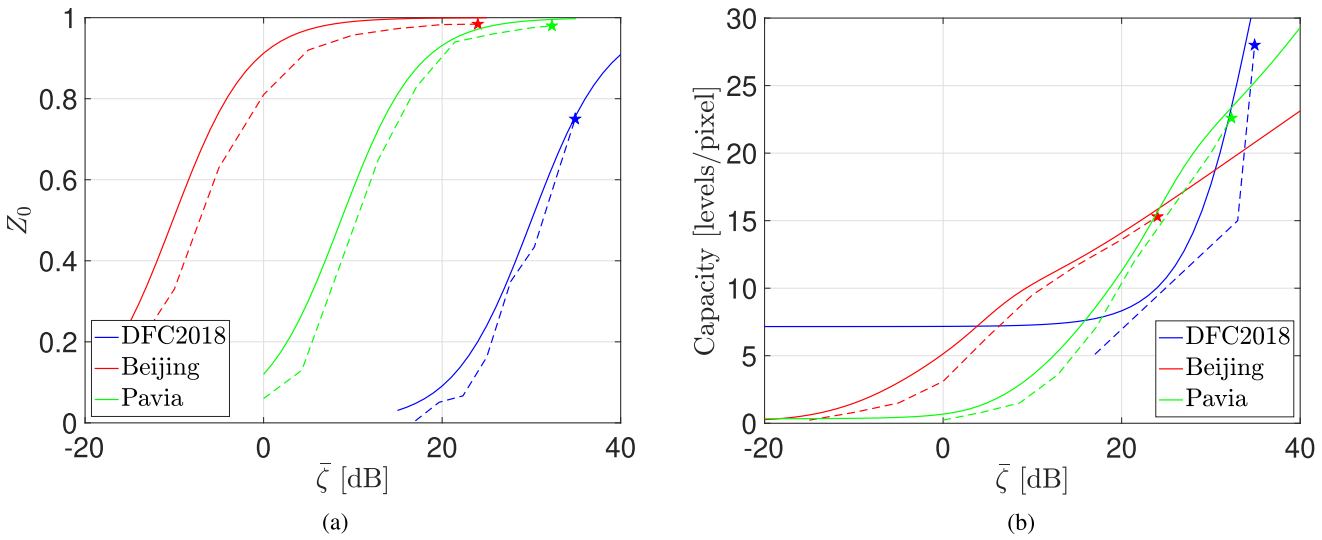


Fig. 5. Reliability and capacity estimates [in (a) and (b), respectively] obtained when considering the IEEE GRSS Data Fusion Contest 2018 data set (blue lines), the Beijing SAR records (red), and the Pavia data (green). The solid lines identify the bounds computed as in (11) and (13). The star markers identify the exact values in (7) and (12) obtained on the actual data sets, whereas the dashed lines show these results obtained on the same data sets where we added white Gaussian noise. These trends are all displayed as a function of $\bar{\zeta}$.

and the goals of the proposed approach. Hence, we can state that the introduced metrics are able to describe the maximum values of reliability and accuracy of the information extraction process for a given multimodal data set under the conditions described in Sections II-A and II-B. Moreover, these figures show that the derived approximate cutoff values are reasonable when $\bar{\zeta}$ is sufficiently large with respect to the actual values of Z_0 computed by the exact equation. Furthermore, they display that although the cutoff estimate in (11) deviates from the true Z_0, it still results in a reasonable capacity estimate approximation. It is also worth noting that the approximate and exact trends are typically following the same behavior for both Z_0 and $\mathcal{C}$ on any considered data sets. We would like to emphasize that the reasons why the DFC2018 data set achieves the highest capacity even though it has the lowest reliability are merely related to the different multimodal setup. As reported in Figs. 2 and 3, the reliability decreases with the number of classes/clusters (equal to 20 for DFC2018, 2 for Beijing, and 5 for Pavia data sets), contrary to the capacity.

C. $\widehat{\mathcal{C}}$ and Overall Accuracy

Let us now compare the capacity index and OA of classification outcomes on the Beijing and Pavia data sets. For each data set, let us randomly identify a subset of M' images. Especially, M' is set to $\{5, 15, 25\}$ for the Beijing data set, while $M' = \{3, 13, 23\}$ when the Pavia data set is considered. To these subsets, we added noise as in the previous experiment so as to obtain different values of $\bar{\zeta}$. In fact, the value of $\bar{\zeta}$ is equal to $\{2, 10, 20\}$ (in dB) when we considered the Beijing data sets, while $\bar{\zeta} = \{5, 10, 20\}$ (in dB) for the Pavia data set. Hence, we take into account nine subsets for both data sets characterized by a different setup of the $(M', \bar{\zeta})$ parameters. Let us now perform classification on these sets of records. When considering the Beijing data set, we are interested in identifying the urbanized area in the region. On the other hand, when we focus our attention on the Pavia records, we aim at understanding the level of air quality of each pixel. Then, let us compute the OA of the resulting classification according to ground-truth maps acquired by ground measurements, as well as the estimated value of capacity according to (13). The OA is obtained over 100 experiments, for each setup $(M', \bar{\zeta})$, with different random subsets of images. In order to provide a fair comparison, we took into account only one classifier, especially the one introduced in [48]. It is worth recalling that the OA achieved by this classifier on the original data set (i.e., without additive noise and using all images so that $(\bar{\zeta}, R, M) = (24, 3, 53)$ for the Beijing data set and $(\bar{\zeta}, R, M) = (32, 5, 35)$ for the Pavia data set) is equal to 94% and 90.6%, respectively. Tables II and III report the results on the aforementioned subsets.

When considering OA, it is typically expected that it would increase for a given value of SNR when the number of images (M') to be processed would increase as well. The results show that this effect is confirmed for higher values of SNR (e.g., $\bar{\zeta} = 20$ dB for the Beijing data set), while the trend becomes the opposite for low-SNR setup (i.e., $\bar{\zeta} = 2$ dB for the Beijing data set). According to Tables II and III, the capacity index proposed in (13) is able to give an indication of this behavior.

The capacity index is especially useful to detect the inversion of the trend in OA in the low-SNR situation, where processing a fewer number of images delivers better results than performing classification with more scenes. According to these results, it is possible to confidently state that the assumptions taken into account in the system setup and throughout the proposed information theory-based analysis are accurate and match the actual data processing structure so that operational use of the proposed indices can be actually developed and implemented.

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TABLE II

OA AND CAPACITY $\widehat{C}$ OBTAINED WHEN CONSIDERING DIFFERENT SUBSETS OF IMAGES (M') AND VALUES OF $\bar{\zeta}$ IN dB ON THE BEIJING DATA SET. OA IS COMPUTED ACCORDING TO THE CLASSIFICATION PERFORMED BY MEANS OF THE METHOD PROPOSED IN [48], WHILE THE CAPACITY IS COMPUTED AS IN (13). μ AND STD REPRESENT THE AVERAGE AND STANDARD DEVIATION, RESPECTIVELY, OF THE OA OBTAINED OVER 100 EXPERIMENTS WITH MULTIPLE RANDOM SUBSETS OF IMAGES

	$M'=5$			$M'=15$			$M'=25$		
	OA			OA			OA		
$\bar{\zeta}$	μ	std	$\widehat{C}$	μ	std	$\widehat{C}$	μ	std	$\widehat{C}$
2	**0.61**	0.04	**2.5**	0.53	0.045	2.1	0.52	0.06	2.1
10	0.71	0.052	5.8	0.79	0.031	7.8	**0.86**	0.036	**8.2**
20	0.74	0.057	9.6	0.85	0.021	11.3	**0.88**	0.013	**12.1**

TABLE III

OA AND CAPACITY $\widehat{C}$ OBTAINED WHEN CONSIDERING DIFFERENT SUBSETS OF IMAGES (M') AND VALUES OF $\bar{\zeta}$ IN dB ON THE PAVIA DATA SET. THE SAME LEGEND AS IN TABLE II APPLIES HERE

	$M'=3$			$M'=13$			$M'=23$		
	OA			OA			OA		
$\bar{\zeta}$	μ	std	$\widehat{C}$	μ	std	$\widehat{C}$	μ	std	$\widehat{C}$
5	**0.62**	0.049	**4.8**	0.52	0.048	2.3	0.51	0.045	2.2
10	**0.69**	0.015	**5**	0.67	0.014	4.95	0.67	0.0143	4.95
20	0.73	0.016	5.3	0.79	0.013	10.1	**0.81**	0.01	**11.3**

D. Improved Classification by Enhanced Automatic Image Selection

To validate the effectiveness of using the capacity metric for automatic selection of the most informative images in a multimodal remote sensing data set, we performed several tests on the records presented in Section IV-A. Especially, we implemented the OBB- and GA-based image selection as introduced in Section III: throughout this section, these methods will be called $\widehat{C}$-OBB and $\widehat{C}$-GA, respectively.

In order to evaluate the actual performance improvement induced by the use of the $\widehat{C}$ metric and understanding the relevance of the images, we performed automatic selection by means of other algorithms widely used in remote sensing technical literature based on different strategies and philosophy. Hence, we considered the following architectures:

1) OBB and GA algorithms when the objective function was set according to the Mahalanobis criterion [37], [42], [46];
2) forward FS based on the one-nearest neighbor leave-one-out criterion [50];
3) principal component analysis (PCA—see, for instance, [51]), based on eigenvalue analysis and ranking;
4) discriminant boundary feature extraction (DBFE) [52], based on separability enhancement;
5) the Fisher information maximization (Fisher information scoring—FIS [53]).

These schemes are generally used at data level for multimodal remote sensing processing.

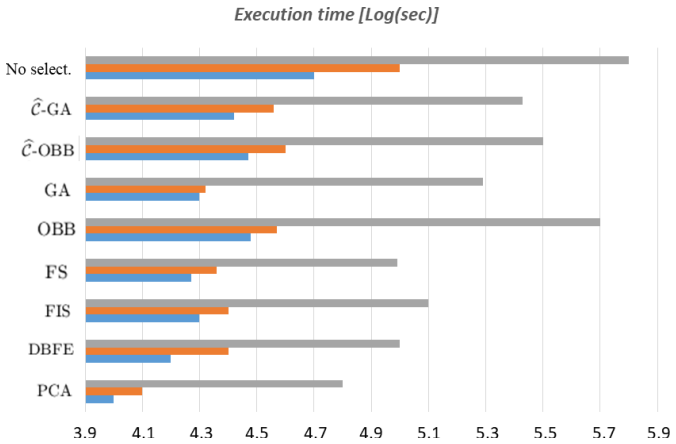


Fig. 6. Execution time (in log(seconds)) required by the architectures in Tables V–XIV (image selection and RF) to achieve the classification of Beijing, Pavia, and IEEE GRSS Data Fusion Contest 2018 data sets (blue, orange, and gray bars, respectively).

All these algorithms are then compared in terms of classification accuracy. In order to provide a fair comparison of the different schemes, we implemented a common platform for classification based on two algorithms, support vector machine (SVM), and random forest (RF), widely used by the remote sensing community (see, for instance, [54] and [55]). This step is meant, on the one hand, to provide a shared platform for the aforementioned schemes of image selection so that consistent evaluation of the performance can be conducted; on the other hand, it gives also the opportunity to quantitatively address the level of applicability of the image selection architectures to different frameworks of remote sensing image classification. When RF is employed, the number of decision trees was empirically set to 100 decision trees, and $\lfloor\sqrt{F}\rfloor^2$ variables are randomly drawn at each node of the trees, where F is the number of images associated with each pixel that has been identified by the aforementioned selection architectures [56], [57]. SVMs exploit radial basis functions [58], with parameters C (penalty parameter of error term) and γ (kernel coefficient) automatically selected via cross-validated grid-search optimization [18].

In summary, for each data set, we have produced eight feature subsets identified by means of methods based on PCA, DBFE, FIS, FS, OBB, and GA at the data level, as well as by the two proposed schemes $\widehat{C}$-OBB and $\widehat{C}$-GA introduced in Section III. Moreover, we have used two different classifiers to evaluate the accuracy performance with respect to the classification of these methods.

1) Beijing Data Set: Let us focus first on the results obtained by processing the Beijing data set. Since the classification problem on this data set is essentially binary, we can evaluate the performances of the different methods according to the performance indices listed in Table IV, where the counts of true positives, true negatives, false positives, and false negatives are listed as TP, TN, FP, and FN, respectively. Table V reports the results achieved. It is possible to appreciate how the

[2] $\lfloor x \rfloor$ denotes the floor function that returns the integer part of a real number x such that $\lfloor x \rfloor \leq x$.

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TABLE IV

DEFINITION OF THE CLASSIFICATION PERFORMANCE INDICES

Index	Acronym	Definition
Accuracy	ACC	$\frac{TP+TN}{TP+FN+TN+FP}$
Sensitivity	TPR	$\frac{TP}{TP+FN}$
Specificity	TNR	$\frac{TN}{FP+TN}$
Precision	PPV	$\frac{TP}{TP+FP}$
F1 score	F1	$\frac{2TP}{2TP+FP+FN}$

TABLE V

QUANTITATIVE ANALYSIS OF THE TESTS CONDUCTED ON THE BEIJING DATA SET. FOR EACH INDEX, THE BEST RESULT IS SHOWN IN RED, WHILE THE SECOND IS SHOWN IN BLUE. M_* REPRESENTS THE OPTIMAL NUMBER OF SELECTED ATTRIBUTES BY EACH METHOD

Method	M_*	Classifier	ACC	TPR	TNR	PPV	F1
PCA	33	RF	0.83	0.87	0.84	0.85	0.86
		SVM	0.83	0.83	0.83	0.88	0.85
DBFE	34	RF	0.82	0.85	0.78	0.84	0.84
		SVM	0.84	0.84	0.84	0.77	0.85
FIS	**26**	RF	0.82	0.87	0.75	0.84	0.85
		SVM	0.82	0.82	0.82	0.88	0.85
FS	30	RF	0.8	0.77	0.79	0.78	0.81
		SVM	0.81	0.81	0.77	0.79	0.81
OBB	35	RF	0.76	0.8	0.79	0.78	0.79
		SVM	0.8	0.78	0.76	0.8	0.8
GA	34	RF	0.79	0.76	0.75	0.79	0.81
		SVM	0.74	0.74	0.8	0.76	0.74
$\widehat{\mathcal{C}}$-OBB	**27**	RF	**0.95**	**0.95**	**0.95**	**0.98**	**0.97**
		SVM	0.92	0.92	0.92	**0.98**	**0.95**
$\widehat{\mathcal{C}}$-GA	**26**	RF	0.91	0.93	0.87	0.95	0.94
		SVM	**0.94**	**0.94**	**0.94**	**0.97**	**0.95**
No slct.	53	RF	0.92	**0.95**	0.88	0.94	0.94

image selection based on the capacity metric (either conducted according to OBB or GA) provides a strong enhancement in all the accuracy indices with the least number of selected images.

To give a complete quantitative assessment of the actual impact of the results, we have reported the results achieved when analyzing the Beijing data set without performing any selection on the records ("No slct." in Table V), as in [48]. The analysis conducted on the whole data set might be more accurate than the investigation performed when using the image selection algorithms. However, the computational cost associated with this scheme is very high with respect to the reduction delivered by the image selection, as it is clear from Fig. 6. Thus, the search for informative records would save the classifier from investigating nonrelevant data, leading to an adequate and effective improvement of the system efficiency. This can be considered as a measure of how the proposed approach can actually guide the decision in ensemble learning and ensemble pruning classification: this effect will be properly explored in future works.

In order to understand how the choice of the images to be processed can actually impact the OA of the classification,

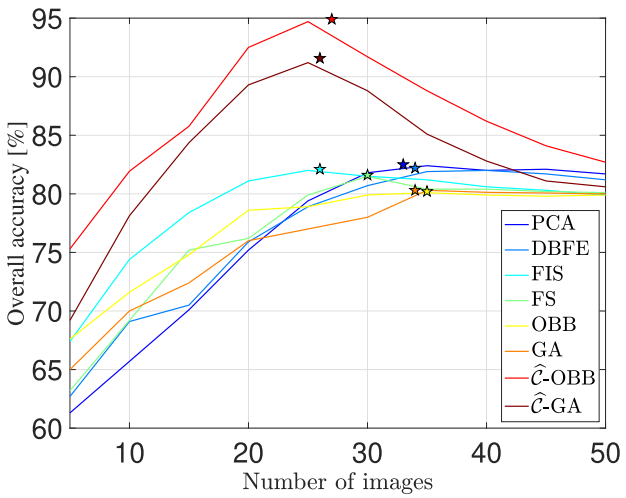


Fig. 7. OA achieved by RF classification as a function of the number of selected images by the different schemes on the Beijing data set. The star markers identify the OA obtained by RF classification when the selection architectures are left free to determine the best set of images (corresponding to results shown in Table V).

we have conducted additional tests on the image selection schemes outcomes. Especially, we have forced the five considered image selection architectures to extract the most relevant images when their number was set to $\{q \cdot 5\}_{q=1,\ldots,10}$. Then, we have computed the OA achieved by means of the RF algorithm. Fig. 7 reports the results. The search for the most informative images can be strongly affected by the number of images that are selected. Indeed, the curves in Fig. 7 show that the accuracy increases as the number of images climbs up to the optimal amount identified by the selection architectures. The accuracy reaches a weak steady behavior as the number of images keeps increasing after this point when selection algorithms based on geometrical and statistical criteria are employed (i.e., PCA, DBFE, and FIS). On the contrary, the accuracy degrades when the number of images passes the optimal selection point defined by the schemes in Section III. This effect shows how the search for most relevant images can be very complicated and possibly mislead by improper setup of the optimization protocols. Moreover, Fig. 7 shows that although FIS and $\widehat{\mathcal{C}}$-GA have the same optimal number of images, our approach achieves a better accuracy that emphasizes the precision of our method at selecting the relevant attributes.

From the previous results, it is possible to notice how the ability to describe the considered multimodal scene depends on the ability to extract relevant features from the data set. It is, thus, important to assess the performance of the different methods to actually pick informative attributes from the set of records to be processed. To this aim, we conducted additional tests on enriched versions of the Beijing data set. Especially, we created 100 data sets where each pixel was characterized by the 53 attributes (corresponding to the $M = 53$ images) that were originally collected in the Beijing data set, plus other 53 fake attributes that were randomly generated by a Gaussian noise having a variance such that the overall SNR for each pixel would be equal to $\{10, 20, 30\}$ dB. According to the

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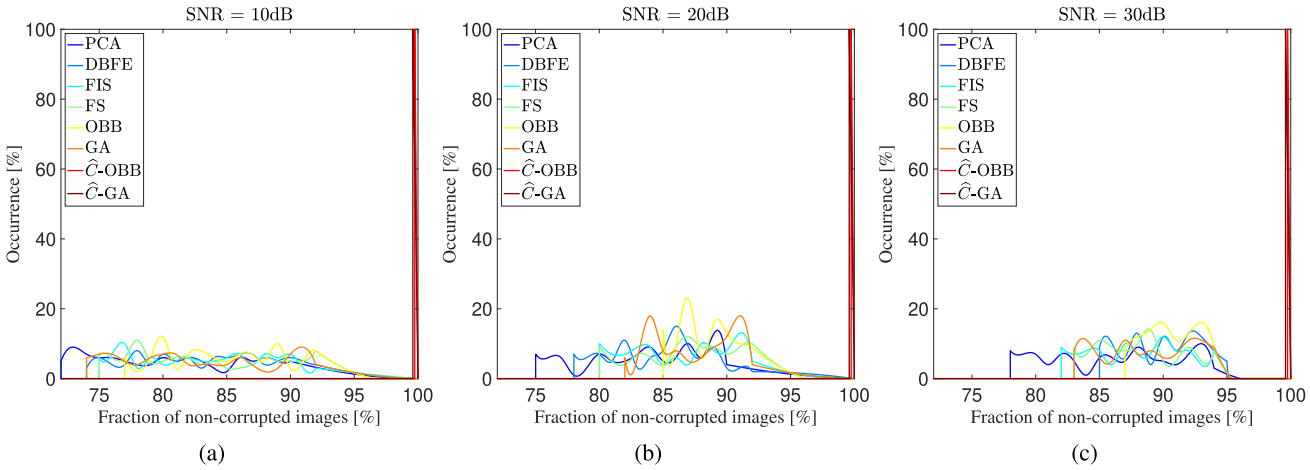


Fig. 8. Distribution of the fraction of noncorrupted images that are selected by the algorithms in Section IV-D when the synthetic data set obtained by corrupting the Beijing data set with noise at SNR set to 10, 20, and 30 dB [(a)–(c), respectively] is considered. We note that our two methods $\widehat{C}$-OBB and $\widehat{C}$-GA only selected relevant images, contrary to other methods, which selected a significant percentage of fake images.

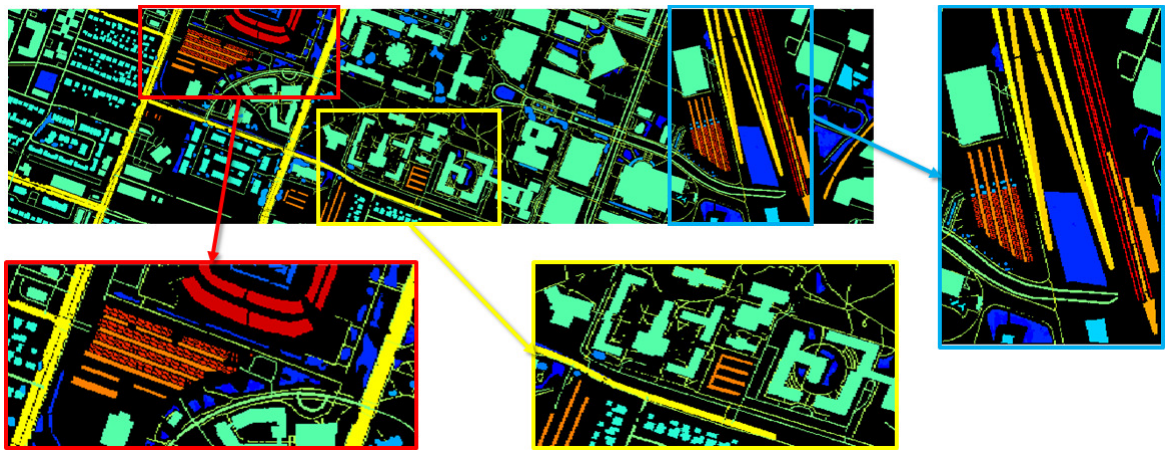


Fig. 9. Ground truth of the 20 classes identified in the IEEE GRSS Data Fusion Contest 2018.

notation that we used throughout this article, for every value of SNR, we had 100 data sets of $M = 106$ images (53 of them being fake images) of 3265×2448 pixels. We used the algorithms that were previously introduced in Section III to select the significant features.

In this case, our attention was focused on understanding whether each method was able to identify the actual attributes that were originally stored in the Beijing data set and to discriminate them from the fake attributes that we added. Hence, we computed, for each SNR, the fraction of noncorrupted images that were selected by every method and calculated the occurrence of this quantity over all the data sets that we created. Fig. 8 summarizes these results for the SNR $= \{10, 20, 30\}$ dB cases [(a)–(c), respectively]. It is possible to appreciate how the proposed $\widehat{C}$-OBB and $\widehat{C}$-GA methods are actually able to discriminate the real images in the Beijing data set from the noisy features that we added. The performance of each other algorithm is suffering from a remarkable percentage of wrong selection of features that can reach up to 25% of the total FS in some cases. This result emphasizes how the proposed approach is able to identify the most informative features in each data set and quantifies the gap between the proposed architectures and the algorithms in the technical literature in retrieving relevant features from the considered scenes.

2) Pavia Data Set: Let us now consider the analysis of the Pavia data set. In this problem, we aim at classifying the 2800×2800 pixels in the scene in five different classes of air quality (i.e., bad, poor, fair, good, and excellent) according to the regulations of the local environmental protection agency. Hence, we are able to derive the confusion matrices for every combination of image selection and classifier. This allows obtaining a more precise characterization of the actual benefits and possible drawbacks of all the methods that we have taken into account. Especially, we can take a look at the confusion matrices that are generated by RF and SVM on the image sets determined by the aforementioned selection algorithms. Tables VI–XIII report these results: they summarize the OA obtained, as well as the producer's accuracy (i.e., the probability that value in a given ground-truth class was classified correctly) and users' accuracy (i.e., the probability that a predicted value actually represents that class in the ground truth) achieved for every air quality class. For a given class, the producer's accuracy, which reflects how well an area has been classified, is measured as the fraction of correctly

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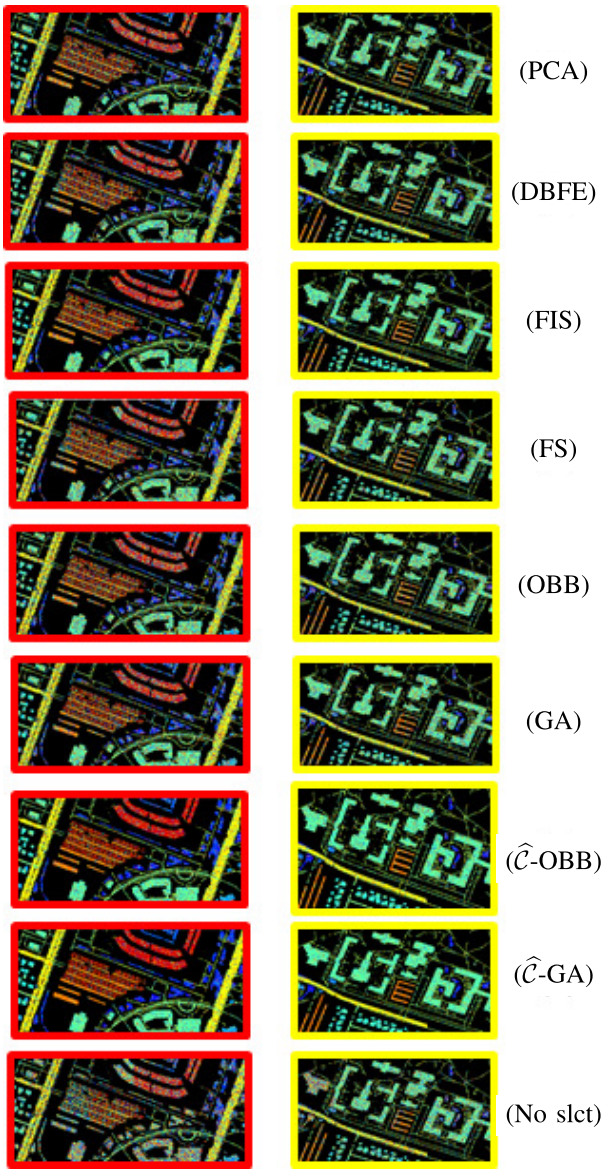


Fig. 10. Details of the classification results on the areas in the red and yellow boxes in Fig. 9. Classification results are achieved by using RF algorithm on the IEEE GRSS Data Fusion Contest 2018 data set when image selection is performed by means of PCA, DBFE, FIS, FS, OBB, GA, $\widehat{\mathcal{C}}$-OBB, and $\widehat{\mathcal{C}}$-GA and when no image selection is performed. The maps show that our approaches ensure a better classification, especially for the stadium seats in red and the roads' class in yellow.

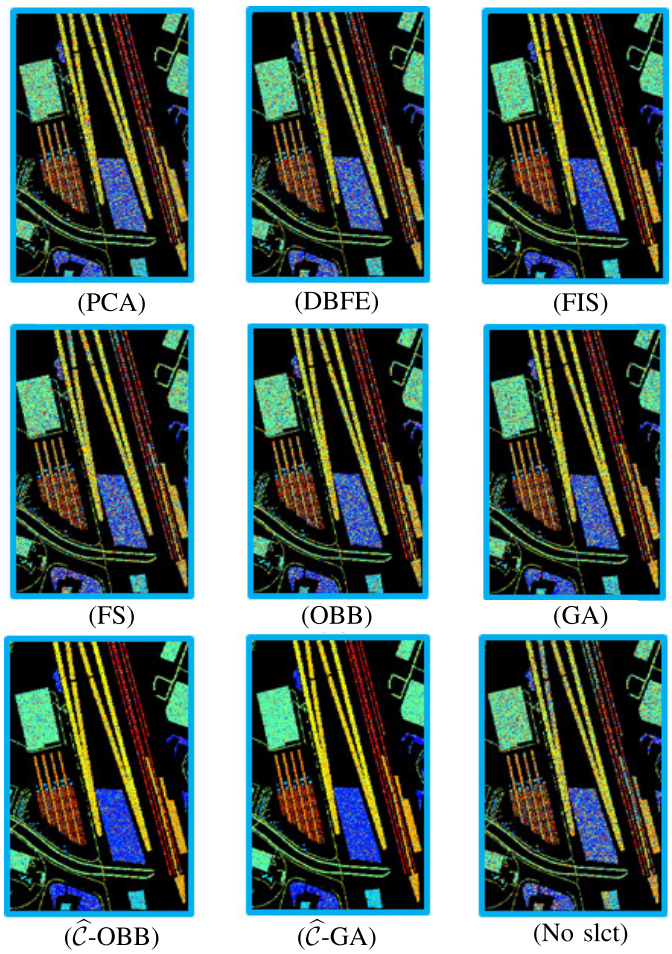


Fig. 11. Details of the classification results on the areas in the light blue box in Fig. 9. Classification results are achieved by using RF algorithm on the IEEE GRSS Data Fusion Contest 2018 data set when image selection is performed by means of PCA, DBFE, FIS, FS, OBB, GA, $\widehat{\mathcal{C}}$-OBB, and $\widehat{\mathcal{C}}$-GA and when no image selection is performed. Our approaches produce better results, especially for the nonresidential buildings in green and the railways in yellow.

predicted values to the total number of ground-truth values, while the users' accuracy, which quantifies the reliability of classification, is calculated as the fraction of correctly predicted values to the total number of classified values [59].

From these tables, it is possible to appreciate how the proposed approach is able to strongly outperform the other methods. In particular, the results obtained for each class display how the information theory-based investigation introduced in this article is able to provide remarkable improvement in characterizing the scenes and their specific components. Indeed, from this analysis on a heterogeneous data set, it is possible to notice that the choice of the classification algorithm does not substantially bias the outcome. Instead, the choice of the selection process is crucial to drive the exploration and reach a solid understanding of the image contents.

In fact, an inadequate reduction of records according to geometrical or statistical properties apparently can jeopardize the investigation of complex data sets (see, for instance, [48], [60], and [6]). Moreover, such reduction might lead to a strong information loss with respect to the analysis conducted without any record selection (see Table XIV) although the efficiency of the system could be enhanced (see Fig. 6). This aspect becomes particularly evident for the Pavia data set (more than in the results of the test on Beijing in Table V), as the records are composed of remote sensing and clinical data. Hence, we can state that the proposed approach identifies a valid solution when investigating the significance of records contained in large-scale and heterogeneous data sets, such that the accuracy of the analysis can be largely improved with an acceptable precision–efficiency tradeoff.

3) IEEE GRSS Data Fusion Contest 2018: Let us finally focus on the IEEE GRSS Data Fusion Contest 2018 [61] data set, whose ground truth is shown in Fig. 9. In this case, in order

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TABLE VI

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 16 IMAGES SELECTED BY PCA OUT OF 35. CLASS COUNTS ($\times 10^3$) ACHIEVED BY USING RF CLASSIFICATION ARE REPORTED: RESULTS OBTAINED BY MEANS OF SVM ANALYSIS ARE DISPLAYED IN BRACKETS. CLASSES FROM GROUND TRUTH AND CLASSIFICATION ARE REPORTED IN COLUMNS AND ROWS, RESPECTIVELY. PA, UA, AND OA RESULTS ARE ALSO REPORTED. FOR EACH INDEX AND EACH CLASS, THE BEST RESULT THAT HAS BEEN ACHIEVED THROUGH THE RESULTS IN TABLES VI–XIV IS SHOWN IN RED, WHILE THE SECOND BEST IS SHOWN IN BLUE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	755.3 (764.2)	187.4 (165.1)	110.9 (99.2)	101.1 (100.5)	66.5 (58.1)	61.8 (64.3)
Poor	182.2 (179.5)	408.1 (413.9)	182.3 (184,5)	213.8 (212.2)	82.4 (80)	38.1 (38.6)
Fair	194.3 (193)	113.5 (122.7)	551.2 (564.5)	512.5 (498.1)	102.3 (96.1)	37.3 (38.2)
Good	178 (176)	42.6 (38.4)	72.6 (79.9)	1410.4 (1432.4)	600.2 (588.8)	61.2 (61.8)
Excellent	101.4 (98.5)	32.4 (43.9)	23.8 (12.7)	506.2 (500.8)	1108.6 (1137)	62.5 (63.4)
PA [%]	53.5 (54.1)	52.05 (52.8)	58.6 (60)	51.4 (52.2)	56.5 (58)	**OA [%]** 54 (55)

TABLE VII

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 20 IMAGES SELECTED BY DBFE OUT OF 35. THE SAME NOTATION OF TABLE VI APPLIES HERE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	778.1 (787.8)	180.7 (158.7)	88.3 (86.2)	54.4 (59.9)	33.3 (45.2)	68.5 (69.2)
Poor	189.9 (185.3)	420.7 (422)	202.1 (201.4)	113.2 (102.2)	40 (40)	43.5 (44.3)
Fair	180.8 (177)	110.6 (122.7)	553 (566)	809.2 (665.2)	52.3 (56.1)	32.4 (35.6)
Good	172.2 (170.1)	40.8 (42.5)	75.2 (66.7)	1420.4 (1446.1)	710.7 (678.4)	58.7 (60.1)
Excellent	90.2 (91)	31.2 (38.1)	22.2 (20.5)	346.8 (470.6)	1123.7 (1140.3)	69.6 (64.7)
PA [%]	55.1 (55.8)	53.6 (53.8)	58.7 (60.1)	51.7 (52.7)	57.3 (58.1)	**OA [%]** 54.7 (55.6)

TABLE VIII

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 17 IMAGES SELECTED BY FIS OUT OF 35. THE SAME NOTATION OF TABLE VI APPLIES HERE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	1002.9 (1040.3)	120.2 (117.9)	42.4 (42.1)	55.2 (60.5)	36.7 (50)	79.7 (79.3)
Poor	290.5 (296.1)	548.8 (555.1)	203.3 (202.9)	100.4 (112.2)	71.7 (68.3)	45.1 (44.9)
Fair	80.4 (48)	55.6 (54.2)	650.8 (651.9)	444.3 (403.6)	123.2 (104.7)	48 (51.6)
Good	20.3 (18.4)	39.3 (36.1)	38.2 (37.7)	1887.8 (1898.8)	348.6 (355.2)	80.8 (80.9)
Excellent	17.1 (8.4)	20.1 (20.7)	6.1 (6.2)	256.3 (268.9)	1379.8 (1381.8)	82.1 (81.9)
PA [%]	71 (73.7)	70 (70.8)	69.2 (69.3)	68.8 (69.2)	70.4 (70.5)	**OA [%]** 69.7 (70.5)

TABLE IX

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 15 IMAGES SELECTED BY FS OUT OF 35. THE SAME NOTATION OF TABLE VI APPLIES HERE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	939.8 (906)	109.3 (111.8)	50.3 (59.8)	150.8 (176.6)	90.1 (87.9)	70.1 (67.5)
Poor	300.1 (285.5)	535.5 (527.6)	202.1 (195)	239 (212.3)	100.2 (155.8)	38.9 (38.3)
Fair	115.2 (123.7)	54.4 (66.2)	589.8 (568.2)	350.4 (372.1)	198.6 (218.1)	45 (42.1)
Good	36 (54.1)	43.7 (45.3)	86.5 (90.5)	1720.5 (1679.3)	244.2 (218.3)	80.7 (80.4)
Excellent	20.1 (41.9)	41.1 (33.1)	12.1 (27.3)	283.3 (303.7)	1326.9 (1279.9)	78.8 (75.9)
PA [%]	66.6 (64.2)	68.3 (67.3)	62.7 (60.4)	61.9 (61.2)	67.7 (65.3)	**OA [%]** 65.2 (63.2)

to highlight the ability of the proposed approach to work at different levels of image fusion, we selected the most relevant bands within the hyperspectral and LiDAR records. Hence, we worked on the whole data cube and run the selection on all the records associated with each pixel. Then, we ran a classification based on RF and SVM algorithms.

Figs. 10 and 11 show the classification maps obtained on this data set. Moreover, Table XV reports the OA and

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TABLE X

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 18 IMAGES SELECTED BY OBB OUT OF 35. THE SAME NOTATION OF TABLE VI APPLIES HERE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	1020.3 (1035.8)	106.7 (92.3)	42.1 (33)	33.5 (6)	46.7 (65.6)	81.6 (84)
Poor	211.2 (190.3)	540.1 (595.8)	134.3 (102.3)	117.3 (86.3)	80.7 (118.8)	49.8 (21.2)
Fair	86.7 (81.1)	72.3 (54.1)	607.7 (611.5)	400.2 (410.3)	243.5 (140.8)	43 (47.1)
Good	65.8 (52.3)	41.7 (25.6)	130.1 (145.4)	1655.9 (1811)	250.4 (282.4)	77.2 (78.1)
Excellent	27.2 (51.7)	23.2 (16.2)	26.6 (48.6)	537.1 (430.4)	1338.7 (1352.4)	68.5 (71.2)
PA [%]	72.3 (73.4)	74.5 (76)	64.6 (65)	60.2 (66)	68.3 (69)	**OA [%]** 65.8 (68.9)

TABLE XI

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 15 IMAGES SELECTED BY GA OUT OF 35. THE SAME NOTATION OF TABLE VI APPLIES HERE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	1030.2 (1059.8)	105.3 (96.3)	39.5 (28.3)	56.4 (45.5)	72.3 (61.4)	79 (82.1)
Poor	197.3 (166.2)	581.7 (588.8)	149.9 (152.8)	100.7 (124)	159.5 (99.9)	48.9 (52)
Fair	96.1 (88.4)	59.5 (74.2)	651 (639.7)	340.2 (256.3)	174.3 (168.2)	49.2 (52.1)
Good	57.3 (69)	32 (16.3)	53.2 (85.8)	1767.1 (1830.2)	287.8 (252.6)	80.4 (78.4)
Excellent	30.3 (27.8)	5.5 (8.4)	47.2 (34.2)	479.6 (488)	1266.1 (1377.9)	69.2 (71.1)
PA [%]	73 (75.1)	74.2 (75.1)	69.2 (68)	64.4 (66.7)	64.6 (70.3)	**OA [%]** 67.5 (70.1)

TABLE XII

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 15 IMAGES SELECTED BY $\widehat{\mathcal{C}}$-OBB OUT OF 35. THE SAME NOTATION OF TABLE VI APPLIES HERE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	1303.9 (1322.3)	25.2 (24.8)	9.8 (8.7)	21.6 (22.3)	10.7 (8.3)	**95** (**95.3**)
Poor	65.3 (63.1)	712.6 (714.2)	45.6 (45.1)	48.3 (44.4)	23 (22.7)	**79.6** (**80.3**)
Fair	24.4 (16.3)	23.4 (21.9)	845.7 (846.7)	57.6 (54.2)	34.2 (32.3)	**85.8** (**87.1**)
Good	10.3 (6.5)	16.7 (15.2)	30.3 (29.8)	2532.7 (2538.2)	69.3 (66.1)	**95.2** (**95.5**)
Excellent	7.3 (3)	6.1 (7.9)	9.4 (10.5)	83.8 (84.9)	1822.8 (1830.6)	**94.4** (**94.5**)
PA [%]	**92.4** (**93.7**)	**90.9** (**91.1**)	**89.9** (**90**)	**92.3** (**92.5**)	**93** (**93.4**)	**OA [%]** **92** (**92.5**)

TABLE XIII

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 14 IMAGES SELECTED BY $\widehat{\mathcal{C}}$-GA OUT OF 35. THE SAME NOTATION OF TABLE VI APPLIES HERE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	1315.2 (1327.9)	22.5 (21.3)	14.6 (13.9)	21.8 (19.7)	12.4 (13.5)	**94.8** (**95.1**)
Poor	52.3 (46.5)	719.7 (722)	26.6 (25.8)	42.9 (40.3)	20.9 (21.4)	**83.4** (**84.3**)
Fair	29.2 (24.4)	22.6 (20.8)	865.5 (867.4)	53.2 (51)	35.3 (33.5)	**86** (**87**)
Good	11.7 (10)	13.4 (12.9)	24.3 (22.2)	2535.4 (2540.9)	72.6 (68.8)	**95.4** (**95.7**)
Excellent	2.8 (2.4)	5.8 (7)	9.8 (11.5)	90.7 (92.1)	1818.8 (1822.8)	**94.3** (**94.1**)
PA [%]	**93.2** (**94.1**)	**91.8** (**92.1**)	**92** (**92.2**)	**92.4** (**92.6**)	**92.8** (**93**)	**OA [%]** **92.5** (**92.8**)

kappa statistics results. In particular, the kappa coefficient measures the agreement between classification and truth values (taking into account the probability of correct classification by chance as well), and it is computed as $K = ((N\sum_{i=1}^{R} A_{ii} - \sum_{i=1}^{R} B_i C_i)/(N^2 - \sum_{i=1}^{R} B_i C_i)) \in [-1, 1]$, where N is the total number of classified values compared with truth values; A_{ii} is the number of values belonging to the truth class i that has also been classified as class i; B_i is the total number of truth values belonging to class i; and C_i is the total number of predicted values belonging to class i. A maximum value of kappa statistics $K = 1$ reflects a perfect agreement between ground truth and classification, while 0

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TABLE XIV

CONFUSION MATRIX OBTAINED ON THE PAVIA DATA SET USING 35 IMAGES WHEN IMAGE SELECTION IS NOT PERFORMED. THE SAME NOTATION OF TABLE VI APPLIES HERE

Air quality class	Bad	Poor	Fair	Good	Excellent	UA [%]
Bad	1032.9 (1031.5)	98.2 (89.9)	35.2 (36.9)	75.6 (68.9)	24.7 (47.4)	81.5 (80.9)
Poor	210.6 (231.5)	537 (518.2)	78.8 (86.5)	183.2 (153.4)	82.4 (77.1)	49.1 (48.5)
Fair	100.3 (92.4)	80.8 (85.3)	660.4 (661.3)	373.8 (366.3)	155.7 (161.1)	48.1 (48.3)
Good	54.8 (36.6)	57.8 (52.3)	106.1 (111.4)	1876.8 (1901.5)	323.3 (302.4)	77.5 (79.1)
Excellent	12.6 (19.2)	10.2 (38.3)	60.3 (44.7)	234.6 (253.9)	1373.9 (1372)	81.2 (79.4)
PA [%]	73.2 (73.1)	68.5 (66.1)	70.1 (70.3)	68.4 (69.3)	70.1 (70)	**OA [%]** 69.9 (69.9)

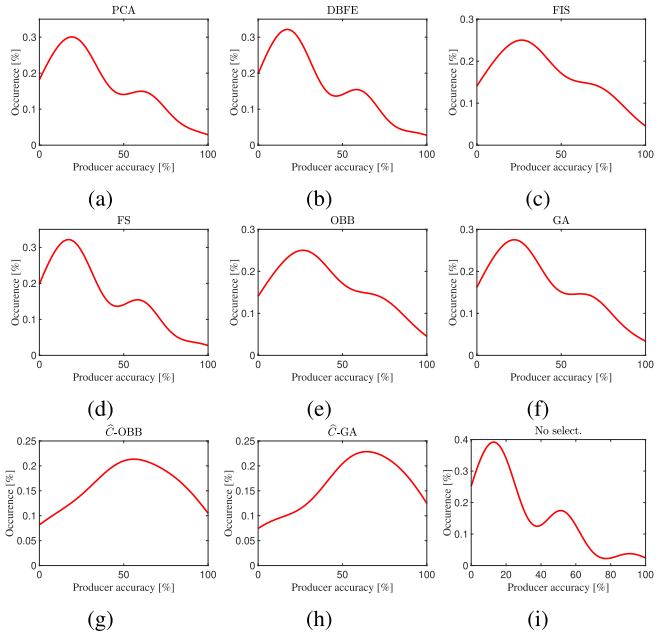


Fig. 12. Producer's accuracy distributions for classification results achieved by using RF algorithm on the IEEE GRSS Data Fusion Contest 2018 data set when image selection is performed by means of PCA, DBFE, FIS, FS, OBB, GA, $\widehat{\mathcal{C}}$-OBB, and $\widehat{\mathcal{C}}$-GA [(a)–(h), respectively] and when no image selection is performed (i). In contrast to our approaches (g) and (h), the other methods show a similar trend with a high concentration toward small values of the producer's accuracy, which reflects their instability.

TABLE XV

ACCURACY ANALYSIS—IEEE GRSS DATA FUSION CONTEST 2018 DATA: OA AND KAPPA STATISTICS (K). FOR EACH INDEX, THE BEST RESULT IS SHOWN IN RED, WHILE THE SECOND IS SHOWN IN BLUE. FOR CLARITY, THE VALUES OF KAPPA STATISTICS WERE MULTIPLIED BY 100. M_* REPRESENTS THE OPTIMAL NUMBER OF SELECTED ATTRIBUTES BY EACH METHOD OUT OF 51

Method	M_*	Classifier	OA [%]	K $\times 100$
PCA	34	RF	51.5	50.9
		SVM	50.7	50.3
DBFE	37	RF	49.7	49.5
		SVM	49.6	47.8
FIS	35	RF	60.2	60
		SVM	59.9	59.3
FS	**32**	RF	53.4	54.2
		SVM	53.1	53.9
OBB	33	RF	57.8	56.6
		SVM	58.1	58
GA	**32**	RF	57.2	55.8
		SVM	53.8	57.3
$\widehat{\mathcal{C}}$-OBB	**31**	RF	**76**	**75.9**
		SVM	74.9	73.8
$\widehat{\mathcal{C}}$-GA	**32**	RF	**77.5**	**76.9**
		SVM	75.7	74.3
No select.	51	RF	40	39.8
		SVM	39.7	39.6

value corresponds to a random agreement and negative values to no agreement. It is prominent to recall that reaching the best accuracy classification results (as targeted by the contest scenario [61]) is out of the scope of this article. Nonetheless, the results that we obtained are instrumental to show the importance of a reliability and significance assessment of the data to be processed. In fact, it is possible to notice how the use of the proposed approach for record selection can actually enhance the classification results that can be achieved by means of classic schemes, such as RF and SVM, and reach performance that is comparable with those obtained by more complex and detailed schemes as in [61].

Indeed, considering the aforesaid results, we can state that data reduction could actually improve the characterization of this multimodal data set, provided that the data reduction is done following relevant approaches. Furthermore, among the selection algorithms that we have considered, the proposed capacity-driven approach is able to distinctly outperform the other selection methods based on eigenvalue analysis, separability enhancement, and the Fisher information maximization. This effect is even more emphasized by Fig. 12, where the occurrence distribution of the producer accuracies (PAs) for each class in the data set is displayed for all the methods in Table XV and Figs. 10 and 11. It is worth recalling that the distribution of producer's accuracy reports information on the precision of the given architecture, i.e., on its ability to classify with low variability. Thus, having a distribution concentrated on the right-hand side of the graph would identify a strong precision of the given framework in assigning labels on the given data set. As for Fig. 12, it is possible to observe how the producer's accuracy trend shifts to higher values when $\widehat{\mathcal{C}}$-OBB and $\widehat{\mathcal{C}}$-GA are employed (as expected in an ideal classification), while, for the other methods, the majority of

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the producer's accuracies are concentrated around values in the interval 10%–30%. Nonetheless, it is worth noticing that this performance improvement translates in a higher computational cost, as reported in Fig. 6, which shows how the execution time of the schemes basically follows the trends already commented for the Beijing and Pavia data sets.

V. Conclusion

In this article, the limits of multimodal remote sensing data analysis in terms of the maximum reliability and accuracy under optimal conditions are investigated. Our approach leads to the definition of two indices that can be easily computed before the actual processing would take place. Thus, the main contributions of this article are as follows:

1) the introduction of an information-theory-based approach to characterize the maximum information extraction performance as a function of data statistics and parameters that characterize any multimodal scenario;
2) the definition of two indices that quantify the maximum accuracy and reliability of the analysis that can take place over the considered multimodal records;
3) asymptotic approximations of the aforementioned indices so that these quantities can be estimated during preprocessing, avoiding the use of difficult numerical integration and root finding techniques;
4) large experimental validation that shows the consistency of our approach, as well as the validity of the proposed indices.

The experimental results show that image selection performed by means of schemes that are not properly addressing informativity maximization might prevent the data analysis scheme from reaching the optimal interpretation of the records and, in some cases, might even degrade the investigation with respect to a scenario where no data reduction is applied. This emphasizes the urgency of a proper assessment of reliability in remote sensing data analysis and especially in multimodal remote sensing investigation so that a precise understanding and interpretation of the Earth's surface phenomena can be retrieved.

For the sake of clarity, we have introduced the reliability and capacity metrics for decision-level fusion. Future works will be devoted to the extension of the results to feature-level fusion. Nevertheless, the development and implementation of platforms that are able to perform enhanced multimodal remote sensing data analysis by considering the proposed metrics during preprocessing for automatic image selection have been presented in this article. Large-scale efficient dimensionality reduction and near-real-time information extraction will be the target of further works.

To accurately assess the capacity and reliability of highly nonlinear approaches of data fusion, such as the ones based on deep learning, a further improvement of the proposed model could be examined by considering spatially correlated noises and nonlinear imperfections. Moreover, multimodal remote sensing analysis is subject to many mismatches due to the heterogeneity of data, such as an improper coregistration or alignment. An interesting perspective of this work would be to investigate how such mismatches could affect the capacity of the analysis to extract information.

Furthermore, by following the approach that we have proposed in this article, it would be possible to assess the actual importance of different modalities in the characterization of specific regions, as a result of the impact of different illumination conditions, registration conditions, and preprocessing operations. Thus, it is possible to expect that this approach might improve the transfer learning performance on multimodal data sets by producing an adaptive scheme to weigh the significance of the different inputs in different areas and, hence, to enhance understanding the physical phenomena.

Appendix A
Computation of the Capacity

The capacity of the system in (1) is defined as the mutual information between $\underline{\underline{\widehat{Y}}}$ and $\underline{\underline{\widehat{X}}}$, maximized over the input distribution

$$\mathcal{C} = \max_{p(\underline{\underline{X}})} I(\underline{\underline{\widehat{X}}}; \underline{\underline{\widehat{Y}}}) = \max_{p(\underline{\underline{X}})} \sum_{l=1}^{L} I(\underline{\hat{x}}_l; \underline{\hat{y}}_l(\underline{\underline{H}}_l)) \tag{16}$$

$$= \max_{p(\underline{\underline{X}})} \sum_{l=1}^{L} \mathbb{E}_{\underline{\underline{H}}_l}[I(\underline{\hat{x}}_l; \underline{\hat{y}}_l|\underline{\underline{H}}_l)]. \tag{17}$$

The distribution of the input $\underline{\hat{x}}_l$ that maximizes $\mathcal{C}$ is Gaussian with covariance $\mathbb{E}[\underline{\hat{x}}_l \underline{\hat{x}}_l^T]$ ($\underline{\hat{x}}_l$ is assumed to be zero-mean) [13], [14]. Using the Gaussian assumptions [62], it follows that:

$$\mathcal{C} = \max_{\hat{P}_l} \sum_{l=1}^{L} \mathbb{E}_{\underline{\underline{H}}_l} \left[\log_Q \left| \underline{\underline{I}}_R + \frac{\underline{\underline{H}}_l \mathbb{E}[\underline{\hat{x}}_l \underline{\hat{x}}_l^T] \underline{\underline{H}}_l^T}{N_0} \right| \right] \tag{18}$$

where $\hat{P}_l = \mathbb{E}[\underline{\hat{x}}_l^T \underline{\hat{x}}_l]$, $\log_Q x = \frac{\ln x}{\ln Q}$ and Q refers to the quantization levels of $\underline{\hat{x}}_l$, and $|.|$ is the determinant operator. We can simplify the expression of $\mathcal{C}$ as follows:

$$\sum_{l=1}^{L} \log_Q \left| \underline{\underline{I}}_R + \frac{\underline{\underline{H}}_l \mathbb{E}[\underline{\hat{x}}_l \underline{\hat{x}}_l^T] \underline{\underline{H}}_l^T}{N_0} \right| \tag{19}$$

$$= \sum_{l=1}^{L} \log_Q \left| \underline{\underline{I}}_R + \frac{\underline{\underline{\Lambda}}_l \mathbb{E}[\underline{\tilde{x}}_l \underline{\tilde{x}}_l^T] \underline{\underline{\Lambda}}_l^T}{N_0} \right| \tag{20}$$

$$\leq \sum_{l=1}^{L} \log_Q \prod_{r=1}^{R} \left(1 + \frac{\lambda_{lr} \hat{P}_{lr}}{N_0} \right) \tag{21}$$

$$= \sum_{l=1}^{L} \sum_{r=1}^{R} \log_Q \left(1 + \frac{\lambda_{lr} \hat{P}_{lr}}{N_0} \right) \tag{22}$$

where (20) follows using that $\underline{\underline{H}}_l = \underline{\underline{U}}_l \underline{\underline{\Lambda}}_l \underline{\underline{V}}_l^T$ and (21) follows using Hadamard's inequality. Hadamard's inequality states that the determinant of a positive definite matrix is less than or equal to the product of its diagonal elements.

Considering, in addition, the power constraint $(1/LR) \sum_{l=1}^{L} \sum_{r=1}^{R} \hat{P}_{lr} = \hat{P}$, we define the capacity as

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in [31], as follows:

$$\mathcal{C} = \max_{\hat{P}_{lr}} \sum_{l=1}^{L}\sum_{r=1}^{R} \mathbb{E}_{\zeta_{lr}}\left[\log_Q\left(1+\frac{\zeta_{lr}\hat{P}_{lr}}{\hat{P}}\right)\right]$$
$$\text{s.t.} \sum_{l=1}^{L}\hat{P}_l = \hat{P} \tag{23}$$

where $\zeta_{lr} = ((\lambda_{lr}\hat{P}_{lr})/N_0)$.

Appendix B
Uniqueness of Z_0

Further details can be found in [14], from where the steps of the following proof is inspired. Using the definition of $p_\zeta(\zeta)$ in (8) and (7) becomes

$$\sum_{r=1}^{R}\kappa_r\int_\nu^\infty\left(\frac{1}{\nu}-\frac{1}{\zeta}\right)\psi_\zeta\left[\mathcal{L}_{r-1}^{R(M-1)}(\zeta)\right]^2\mathrm{d}\zeta = RL\bar{\zeta} \tag{24}$$

where $\psi_\zeta = e^{-\zeta}\zeta^{R(M-1)}$. Let us define $\phi_r(\zeta,\nu) = ((1/\nu)-(1/\zeta))e^{-\zeta}\zeta^{R(M-1)}[\mathcal{L}_{r-1}^{R(M-1)}(\zeta)]^2$ [14]. Furthermore, let us consider

$$\Phi(\nu) = \sum_{r=1}^{R}\kappa_r\int_\nu^\infty\phi_r(\zeta,\nu)\mathrm{d}\zeta - RL\bar{\zeta}. \tag{25}$$

Then, it is possible to prove that $\partial\Phi(\nu)/\partial\nu < 0$ and $\partial^2\Phi(\nu)/\partial\nu^2 > 0$ for $\nu > 0$. Moreover, due to the normalization property of the probability distribution function, the following holds:

$$\lim_{\nu\to 0^+}\int_\nu^\infty e^{-\xi}\xi^{R(M-1)}\left[\mathcal{L}_{r-1}^{R(M-1)}(\xi)\right]^2\mathrm{d}\xi = \kappa_r^{-1}. \tag{26}$$

Analogously, we can write as follows, considering $\mathcal{U}(\nu) = \int_\nu^\infty e^{-\xi}\xi^{R(M-1)-1}[\mathcal{L}_{r-1}^{R(M-1)}(\xi)]^2\mathrm{d}\xi$

$$\lim_{\nu\to 0^+}\mathcal{U}(\nu) = \frac{\partial^{r-1}}{\partial w^{r-1}}S(R(M-1),w)\bigg|_{w=0}T(R,M,r) \tag{27}$$

where

$$T(R,M,r) = \frac{\Gamma(R(M-1)\Gamma(R(M-1)+r)}{\Gamma(R(M-1)+1)((r-1)!)^2} \tag{28}$$

being $\Gamma(t) = \int_0^\infty u^{t-1}e^{-t}\mathrm{d}u$. Moreover, $S(R(M-1),w) = F[R(M-1)/2;(R(M-1)+1)/2;R(M-1)+1;4w/(1+w)^2]/[(1-w)(1+w)^{R(M-1)}]$, where

$$F[t_1;t_2;t_3;t_4] = \sum_{j=0}^{\infty}\frac{(t_1)_j(t_2)_j}{(t_3)_j}\frac{t_4^j}{j!} \tag{29}$$

where we have $(v)_j = \prod_{h=0}^{j-1}v+h$, and $(v)_0 = 1$. Thus, by means of an hypergeometric transformation to be applied on (27) according to the aforementioned definitions

$$\lim_{\nu\to 0^+}\mathcal{U}(\nu) = \frac{(R(M-1)+r-1)!}{(R(M-1))(r-1)!}. \tag{30}$$

Therefore, using this result and (26) in the definition of $\Phi(\nu)$, $\lim_{\nu\to 0^+}\Phi(\nu) = +\infty$ and $\lim_{\nu\to\infty}\Phi(\nu) = -RL\bar{\zeta}$.

Thus, since $\partial\Phi(\nu)/\partial\nu < 0$, $\Phi(\nu)$ has a unique value subject to $\Phi(\nu) = 0$. Hence, for any $\bar{\zeta} > 0$, there is a unique Z_0 that satisfies (7).

Appendix C
Details of Z_0 Computation

The term $\mathcal{L}_s^t(u)$ in (9) can be written, using the binomial expansion, as follows:

$$\mathcal{L}_s^t(u) = \sum_{j=0}^{s}(-1)^j\binom{s+t}{s-j}u^j/j!. \tag{31}$$

With this in mind, (24) becomes

$$L\sum_{r=1}^{R}\kappa_r\sum_{j_1=0}^{r-1}\sum_{j_2=0}^{r-1}\frac{(-1)^{j_1+j_2}}{j_1!j_2!}J_1J_2\mathcal{G}_{j_1j_2}(\nu) = RL\bar{\zeta} \tag{32}$$

where $\mathcal{G}_{j_1j_2}(\nu) = \int_\nu^\infty((1/\nu)-(1/\zeta))e^{-\zeta}\zeta^{R(M-1)+j_1+j_2}\mathrm{d}\zeta$, as $j_1+j_2 = 0,1,\ldots,2(R-1)$. Furthermore, since $\int_\nu^\infty e^{-\zeta}\zeta^{RM}\mathrm{d}\zeta = (RM)!e^{-\nu}\sum_{k=0}^{RM}\nu^k/k!$, we can write $\mathcal{G}_{j_1j_2}(\nu) = \Gamma(\beta_1,\nu)/\nu - \Gamma(\beta_2,\nu)$, so (32) can be written as (10).

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References

[1] L. Gomez-Chova, D. Tuia, G. Moser, and G. Camps-Valls, "Multimodal classification of remote sensing images: A review and future directions," *Proc. IEEE*, vol. 103, no. 9, pp. 1560–1584, Sep. 2015.

[2] A. Voisin, V. A. Krylov, G. Moser, S. B. Serpico, and J. Zerubia, "Supervised classification of multisensor and multiresolution remote sensing images with a hierarchical copula-based approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3346–3358, Jun. 2014.

[3] S. Chlaily, C. Ren, P.-O. Amblard, O. Michel, P. Comon, and C. Jutten, "Information–estimation relationship in mismatched Gaussian channels," *IEEE Signal Process. Lett.*, vol. 24, no. 5, pp. 688–692, May 2017.

[4] F. Sedighin, M. Babaie-Zadeh, B. Rivet, and C. Jutten, "Multimodal soft nonnegative matrix co-factorization for convolutive source separation," *IEEE Trans. Signal Process.*, vol. 65, no. 12, pp. 3179–3190, Jun. 2017.

[5] N. Longbotham *et al.*, "Multi-modal change detection, application to the detection of flooded areas: Outcome of the 2009–2010 data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 1, pp. 331–342, Feb. 2012.

[6] M. Datcu *et al.*, "Information mining in remote sensing image archives: System concepts," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 12, pp. 2923–2936, Dec. 2003.

[7] M. Dalla Mura, S. Prasad, F. Pacifici, P. Gamba, J. Chanussot, and J. A. Benediktsson, "Challenges and opportunities of multimodality and data fusion in remote sensing," *Proc. IEEE*, vol. 103, no. 9, pp. 1585–1601, Sep. 2015.

[8] P. Ghamisi *et al.*, "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[9] M. Schmitt and X. X. Zhu, "Data fusion and remote sensing: An ever-growing relationship," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 4, pp. 6–23, Dec. 2016.

[10] D. Lazer, R. Kennedy, G. King, and A. Vespignani, "The parable of Google flu: Traps in big data analysis," *Science*, vol. 343, no. 6176, pp. 1203–1205, Mar. 2014.

[11] D. Mackay, *Information Theory, Inference and Learning Algorithms*. Cambridge, U.K.: Cambridge Univ. Press, 2003.

[12] J. Zhang *et al.*, "Evolutionary computation meets machine learning: A survey," *IEEE Comput. Intell. Mag.*, vol. 6, no. 4, pp. 68–75, Nov. 2011.

[13] E. Biglieri, R. Calderbank, A. Constantinides, A. Goldsmith, A. Paulraj, and H. V. Poor, *MIMO Wireless Communications*. Cambridge, U.K.: Cambridge Univ. Press, 2007.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:23:04 UTC from IEEE Xplore. Restrictions apply.

[14] M.-S. Alouini and A. J. Goldsmith, "Capacity of Rayleigh fading channels under different adaptive transmission and diversity-combining techniques," *IEEE Trans. Veh. Technol.*, vol. 48, no. 4, pp. 1165–1181, Jul. 1999.

[15] A. Marinoni and P. Gamba, "An efficient approach for local affinity pattern detection in remotely sensed big data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 10, pp. 4622–4633, Oct. 2015.

[16] W. Li, Q. Du, and M. Xiong, "Kernel collaborative representation with Tikhonov regularization for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 48–52, Jan. 2015.

[17] G. Camps-Valls, L. Gomez-Chova, K. Richter, and J. Calpe-Maravilla, "Biophysical parameter estimation with a semisupervised support vector machine," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 2, pp. 248–252, Apr. 2009.

[18] G. Camps Valls and L. Bruzzone, *Kernel Methods for Remote Sensing Data Analysis*. Hoboken, NJ, USA: Wiley, 2009.

[19] J. Chen, J. Xia, P. Du, J. Chanussot, Z. Xue, and X. Xie, "Kernel supervised ensemble classifier for the classification of hyperspectral data using few labeled samples," *Remote Sens.*, vol. 8, no. 7, p. 601, Jul. 2016.

[20] B. Song *et al.*, "Remotely sensed image classification using sparse representations of morphological attribute profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 5122–5136, Aug. 2014.

[21] X. Li, S. Zhang, X. Pan, P. Dale, and R. Cropp, "Straight road edge detection from high-resolution remote sensing images based on the ridgelet transform with the revised parallel-beam radon transform," *Int. J. Remote Sens.*, vol. 31, no. 19, pp. 5041–5059, Oct. 2010.

[22] G. Campsvalls, L. Gomezchova, J. Munozmari, J. Vilafrances, J. Amoroslopez, and J. Calpemaravilla, "Retrieval of oceanic chlorophyll concentration with relevance vector machines," *Remote Sens. Environ.*, vol. 105, no. 1, pp. 23–33, Nov. 2006.

[23] K. Song, D. Lu, L. Li, S. Li, Z. Wang, and J. Du, "Remote sensing of chlorophyll-a concentration for drinking water source using genetic algorithms (GA)-partial least square (PLS) modeling," *Ecol. Informat.*, vol. 10, pp. 25–36, Jul. 2012.

[24] H. Nguyen, K. Koike, and M. Nhuan, "Improved accuracy of chlorophyll-a concentration estimates from MODIS imagery using a two-band ratio algorithm and geostatistics: As applied to the monitoring of eutrophication processes over Tien Yen Bay (Northern Vietnam)," *Remote Sens.*, vol. 6, no. 1, pp. 421–442, Dec. 2013.

[25] S. Ihara, "On the capacity of channels with additive non-Gaussian noise," *Inf. Control*, vol. 37, no. 1, pp. 34–39, Apr. 1978.

[26] G. H. Golub and C. F. Van Loan, *Matrix Computations*. Baltimore, MD, USA: Johns Hopkins Univ. Press, 3nd edition, 1996.

[27] G. Blower, *Random Matrices: High Dimensional Phenomena* (London Mathematical Society Lecture Note Series). Cambridge, U.K.: Cambridge Univ. Press, 2009.

[28] J. R. Vergara and P. A. Estévez, "A review of feature selection methods based on mutual information," *Neural Comput. Appl.*, vol. 24, no. 1, pp. 175–186, Jan. 2014.

[29] A. El Gamal and Y.-H. Kim, *Network Information Theory*. Cambridge, U.K.: Cambridge Univ. Press, 2011.

[30] T. W. Anderson, *An Introduction to Multivariate Statistical Analysis* (Wiley Series in Probability and Statistics). Hoboken, NJ, USA: Wiley, 2003.

[31] A. J. Goldsmith and P. P. Varaiya, "Capacity of fading channels with channel side information," *IEEE Trans. Inf. Theory*, vol. 43, no. 6, pp. 1986–1992, 1997.

[32] A. Kraskov, H. Stögbauer, and P. Grassberger, "Estimating mutual information," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 69, no. 6, Jun. 2004, Art. no. 066138.

[33] S. Boyd and L. Vandenberghe, *Convex optimization*. New York, NY, USA: Cambridge Univ. Press, 2004.

[34] S. Chakrabarti, J. Judge, T. Bongiovanni, A. Rangarajan, and S. Ranka, "Disaggregation of remotely sensed soil moisture in heterogeneous landscapes using holistic structure-based models," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4629–4641, Aug. 2016.

[35] D. Tuia, R. Flamary, and M. Barlaud, "To be or not to be convex? A study on regularization in hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4947–4950.

[36] K. Svanberg, "The method of moving asymptotes—A new method for structural optimization," *Int. J. Numer. Methods Eng.*, vol. 24, no. 2, pp. 359–373, 1987.

[37] A. A. Zhigliavsky and J. Pinter, *Theory of Global Random Search*. Dordrecht, The Netherlands: Kluwer, 1991.

[38] R. Horst and H. Tuy, *Global Optimization–Deterministic Approaches*, 3rd ed. Berlin, Germany: Springer, 1996.

[39] W. Forster, "Homotopy methods," in *Handbook of Global Optimization. Nonconvex Optimization and Its Applications*, R. Horst and P.M. Pardalos, Eds. Boston, MA, USA: Springer, 1995.

[40] J. Mockus, W. Eddy, and G. Reklaitis, *Bayesian Heuristic Approach to Discrete and Global Optimization*. Boston, MA, USA: Springer, 1996.

[41] Q. Zheng and D. Zhuang, "Integral global minimization: Algorithms, implementations and numerical tests," *J. Global Optim.*, vol. 7, no. 4, pp. 421–454, Dec. 1995.

[42] S. Voss, S. Martello, I. H. Osman, and C. Roucairol, *Meta-Heuristics: Advances and Trends in Local Search Paradigms for Optimization*. London, U.K.: Kluwer, 1999.

[43] K. A. Dill, A. T. Phillips, and J. B. Rosen, "Molecular structure prediction by global optimization," in *Developments in Global Optimization*, I. M. Bomze, T. Csendes, R. Horst, and P. M. Pardalos, Eds. London, U.K.: Kluwer,1997.

[44] J. J. Moré and Z. Wu, "Global continuation for distance geometry problems," *SIAM J. Optim.*, vol. 7, no. 3, pp. 814–836, Aug. 1997.

[45] A. V. Levy and S. Gomez, "The tunneling method applied for the global minimization of functions," in *Numerical Optimization*, P. T. Boggs, Ed. Philadelphia, PA, USA: SIAM, 1985.

[46] J. M. Fowkes, N. I. M. Gould, and C. L. Farmer, "A branch and bound algorithm for the global optimization of Hessian Lipschitz continuous functions," *J. Global Optim.*, vol. 56, no. 4, pp. 1791–1815, Aug. 2013.

[47] (2018). *IEEE GRSS Data Fusion Contest*. [Online]. Available: http://www.grss-ieee.org/community/technical-committees/data-fusion

[48] A. Marinoni, G. C. Iannelli, and P. Gamba, "An information theory-based scheme for efficient classification of remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5864–5876, Oct. 2017.

[49] A. Dagliati *et al.*, "Integration of administrative, clinical, and environmental data to support the management of type 2 diabetes mellitus: From satellites to clinical care," *J. Diabetes Sci. Technol.*, vol. 10, no. 1, pp. 19–26, 2016.

[50] L. Bruzzone, "An approach to feature selection and classification of remote sensing images based on the bayes rule for minimum cost," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 1, pp. 429–438, Jan. 2000.

[51] R. D. Phillips, L. T. Watson, R. H. Wynne, and C. E. Blinn, "Feature reduction using a singular value decomposition for the iterative guided spectral class rejection hybrid classifier," *ISPRS J. Photogramm. Remote Sens.*, vol. 64, no. 1, pp. 107–116, Jan. 2009.

[52] C. Lee and D. A. Landgrebe, "Feature extraction based on decision boundaries," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 15, no. 4, pp. 388–400, Apr. 1993.

[53] Q. Gu, Z. Li, and J. Han, "Generalized Fisher score for feature selection," in *Proc. 27th Conf. Uncertainty Artif. Intell. (UAI)*, 2012, pp. 266–273.

[54] A. Zafari, R. Zurita-Milla, and E. Izquierdo-Verdiguier, "Evaluating the performance of a random forest kernel for land cover classification," *Remote Sens.*, vol. 11, no. 5, p. 575, Mar. 2019.

[55] J. Xia, N. Falco, J. Benediktsson, P. Du, and J. Chanussot, "Hyperspectral image classification with rotation random forest via KPCA," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1601–1609, Apr. 2017.

[56] C. Strobl, A.-L. Boulesteix, and T. Augustin, "Unbiased split selection for classification trees based on the Gini index," *Comput. Statist. Data Anal.*, vol. 52, no. 1, pp. 483–501, 2007.

[57] T. K. Ho, "A data complexity analysis of comparative advantages of decision forest constructors," *Pattern Anal. Appl.*, vol. 5, no. 2, pp. 102–112, Jun. 2002.

[58] C. J. C. Burges, "A tutorial on support vector machines for pattern recognition," *Data Mining Knowl. Discovery*, vol. 2, no. 2, pp. 121–167, 1998.

[59] S. V. Stehman, "Selecting and interpreting measures of thematic classification accuracy," *Remote Sens. Environ.*, vol. 62, no. 1, pp. 77–89, Oct. 1997.

[60] W. Li and Q. Du, "A survey on representation-based classification and detection in hyperspectral remote sensing imagery," *Pattern Recognit. Lett.*, vol. 83, pp. 115–123, Nov. 2016.

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[61] Y. Xu *et al.*, "Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1709–1724, Jun. 2019.

[62] T. Cover and J. Thomas, *Elements of Information Theory (Wiley Series in Telecommunications and Signal Processing)*. New York, NY, USA: Wiley, 2006.



Saloua Chlaily (Member, IEEE) received the double M.Sc. degrees in electronics engineering and electrical engineering from École Nationale Supérieure d'Électrotechnique, d'Électronique, d'Informatique, d'Hydraulique et des Télécommunications (ENSEEIHT), Toulouse, France, and Hassania School of Public Works (EHTP), Casablanca, Morocco, in 2013, and the Ph.D. degree in signal, image, speech, and telecommunication from Grenoble Alpes University, Grenoble, France, in 2018.

She is a Post-Doctoral Fellow with UiT The Arctic Univesity of Norway, Tromsø, Norway. She is conducting her research at the Center of Integrated Remote Sensing and Forecasting for Arctic Operations (CIRFA). Her research interest includes multimodal data analysis, signal processing, and image processing.

Dr. Chlaily is a Reviewer of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.



Mauro Dalla Mura (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in telecommunication engineering from the University of Trento, Trento, Italy in 2005 and 2007, respectively, the Ph.D. degree in information and communication technologies (telecommunications area) from the University of Trento, and the Ph.D. degree in electrical and computer engineering from the University of Iceland, Iceland, in 2011.

In 2011, he was a Research Fellow with Fondazione Bruno Kessler, Trento, conducting research on computer vision. He has been appointed Specially Appointed Associate Professor with the School of Computing, Tokyo Institute of Technology, Tokyo, Japan, from 2019 to 2022. He is an Assistant Professor with Grenoble Institute of Technology (Grenoble INP), Grenoble, France. He is conducting his research at the Grenoble Images Speech Signals and Automatics Laboratory (GIPSA-Lab). His main research activities are in the fields of remote sensing, image processing and pattern recognition. In particular, his interests include multispectral and hyperspectral image processing, computational imaging, and the analysis of geophysical signals.

Dr. Dalla Mura was a recipient of the IEEE GRSS Second Prize in the Student Paper Competition of the IEEE IGARSS 2011 and a co-recipient of the Best Paper Award of the International Journal of Image and Data Fusion for the year 2012–2013 and the Symposium Paper Award for IEEE IGARSS 2014. He has been the President of the IEEE GRSS French Chapter since 2016 (he previously served as Secretary from 2013 to 2016). In 2017, the IEEE GRSS French Chapter was the recipient of the IEEE GRSS Chapter Award and the Chapter of the year 2017 from the IEEE French Section. He has been on the Editorial Board of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (J-STARS) since 2016.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP, where he is a Professor of signal and image processing. He is a Visiting Scholar with Stanford University, Stanford, CA, USA, KTH, Stockholm, Sweden, and NUS, Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavík, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California, Los Angeles (UCLA), Los Angeles, CA, USA. He holds the AXA Chair in remote sensing and also an Adjunct Professor with the Chinese Academy of Sciences, Aerospace Information Research Institute, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008 and the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He is a member of the Institut Universitaire de France from 2012 to 2017 and a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters, 2018–2019). He has received multiple outstanding paper awards. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote sensing (WHISPERS). He was the Chair from 2009 to 2011 and a Co-Chair of the GRS Data Fusion Technical Committee from 2005 to 2008. He is the Founding President of the IEEE Geoscience and Remote Sensing French chapter from 2007 to 2010 which received the 2010 IEEE GRS-S Chapter Excellence Award. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING and the PROCEEDINGS OF THE IEEE. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he served as a Guest Editor for the *IEEE Signal Processing Magazine*.



Christian Jutten (Fellow, IEEE) received the Ph.D. and Doctor ès Sciences degrees in signal processing from the Grenoble Institute of Technology (GIT), Grenoble, France, 1981 and 1987, respectively.

He has been Visiting Professor with EPFL, Lausanne, Switzerland, Riken labs, Saitama, Japan, and Campinas University, Campinas - SP, Brazil. He served as the Director or Deputy Director with the Signal Processing Laboratory from 1993 to 2010. He was a Scientific Advisor for signal/image processing with the French Ministry of Research from 1996 to 1998 and CNRS since 2012. He was an Associate Professor from 1982 to 1989 and a Full Professor from 1989 to 2019 with University Grenoble Alpes, Grenoble, where he is emeritus Professor. He is the author/coauthor of 4 books, 110+ articles in international journals, 27 keynote plenary talks, and 250+ publications in international conferences. His research interests are in machine learning and source separation, including theory and applications (brain and hyperspectral imaging, chemical sensor array, speech).

Dr. Jutten is a fellow of EURASIP. He was a member of the IEEE MLSP and SPTM Technical Committees. He has been a Senior Member of Institut Universitaire de France since 2008, and the recipient of the 2012 ERC Advanced Grant for the project Challenges in Extraction and Separation of Sources (CHESS). He was an Organizer or Program Chair of many international conferences, especially of the 1st International Conference on Blind Signal Separation and Independent Component Analysis in 1999 (ICA'99) and ICASSP 2020. He received many awards, e.g. Best Paper Awards of EURASIP in 1992 and IEEE GRSS in 2012, Medal Blondel in 1997 from the French Electrical Engineering Society, and one Grand Prix of the French Académie des Sciences in 2016.

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Paolo Gamba (Fellow, IEEE) received the Laurea degree *(cum laude)* and the Ph.D. degree in electronic engineering from the University of Pavia, Pavia, Italy, in 1989 and 1993, respectively.

He is a Professor with the University of Pavia, where he leads the Telecommunications and Remote Sensing Laboratory. He has authored or coauthored more than 140 articles in international peer-review journals and presented nearly 300 research works in workshops and conferences. He served as an Editor-in-Chief for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS from 2009 to 2013, and as the Chair for the Data Fusion Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) from October 2005 to May 2009. He has been elected in the GRSS AdCom since 2014, and served as a GRSS President from 2019 to 2020. He was the organizer and Technical Chair of the biennial GRSS/ISPRS Joint Workshops on Remote Sensing and Data Fusion over Urban Area from 2001 to 2015. He also served as a Technical Co-Chair of the 2010, 2015, and 2020 IGARSS conferences, in Honolulu (Hawaii), Milan (Italy), and Waikoloa (Hawaii), respectively. He has been invited to give keynote lectures and tutorials on several occasions about urban remote sensing, data fusion, EO data for physical exposure, and risk management.



Andrea Marinoni (Senior Member, IEEE) received the B.S., M.Sc. *(cum laude)*, and the Ph.D. degrees in electronic engineering from the University of Pavia, Pavia, Italy, in 2005, 2007, and 2011, respectively.

From 2013 to 2018, he was a Research Fellow with the Telecommunications and Remote Sensing Laboratory, Department of Electrical, Computer and Biomedical Engineering, University of Pavia. In 2009, he was a Visiting Researcher with the Communications Systems Laboratory, Department of Electrical Engineering, University of California, Los Angeles (UCLA), Los Angeles, CA, USA. From 2015 to 2017, he was a Visiting Researcher with Earth and Planetary Image Facility, Ben-Gurion University of the Negev, Be'er Sheva, Israel; School of Geography and Planning, Sun Yat-Sen University, Guangzhou, China; School of Computer Science, Fudan University, Shanghai, China; Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China; Instituto de Telecomunicações, Instituto Superior Tecnico, Universidade de Lisboa, Lisbon, Portugal. In 2020, he was a Visiting Professor with the Department of Electrical, Computer and Biomedical Engineering, University of Pavia. He is an Associate Professor with Earth Observation Group, Center for Integrated Remote Sensing and Forecasting for Arctic Operations (CIRFA), Department of Physics and Technology, UiT the Arctic University of Norway, Tromsø, Norway. His main research interests are focused on efficient information extraction from multimodal remote sensing, nonlinear signal processing applied to large-scale heterogeneous records, Earth observation interpretation and Big Data mining, analysis, and management for human–environment interaction assessment.

Dr. Marinoni has been the recipient of the 2-year Applied Research Grant, sponsored by the Region of Lombardy, Italy, and STMicroelectronics N.V., since 2011. In 2017, he was the recipient of the INROAD Grant, sponsored by the University of Pavia and Fondazione Cariplo, Italy, for supporting excellence in the design of the ERC proposal. In 2018, he was the recipient of the Progetto professionalitá Ivano Becchi Grant funded by Fondazione Banco del Monte di Lombardia, Italy, and sponsored by University of Pavia, and NASA Jet Propulsion Laboratory, Pasadena, CA, for supporting the development of advanced methods of air pollution analysis by remote sensing data investigation. He was the recipient of Åsgard Research Programme and Åsgard Recherche+ Programme grants funded by Institut Français de Norvège, Oslo, Norway, in 2019 and 2020, respectively, for supporting the development of scientific collaborations between French and Norwegian research institutes. He is the Founder and Current Chair of the IEEE GRSS Norway chapter. He is also an Ambassador for the IEEE Region 8 Humanitarian activities, and a Research Contact Point for the Norwegian Artificial Intelligence Research Consortium (NORA—nora.ai). He serves as a Topical Associate Editor for machine learning for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

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